bigT

P2P settlement — CNY rails demo (WeChat / Alipay / Bank)

P2P settlement · WeChat / Alipay / bank
Step-by-step demo guide — the CNY rails
captured 2026-10-10T15:03:03.446Z

Status: **built and captured**. Every screenshot below was taken from the real wallet UI driving the real `p2p-engine` — one engine, no mocks. The three CNY legs were captured by `e2e/capture-p2p-cny.mjs` (a fresh wallet pair per rail). The rail is a data field, not a fork (`docs/p2p.md` §7): the same 2-of-3 escrow, the same PQ-signed state machine, three different fiat legs.

Companion documents: docs/p2p.md (design, protocol & threat model), engine source `services/p2p-engine/src/`.

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0. What this demo proves (read first)

1. **One state machine, three fiat rails.** `wantRail ∈ {wechat, alipay, bank}` decides the whole fiat leg; all three use `instructions` / `proof` / `confirm` / `release` / `complete` (`state.ts:9-28`). Everything before `MATCHED` and the escrow lock is identical.
2. **2-of-3 P2SH escrow on Bigtangle L0.** Seller, buyer and engine hold one key each; any two satisfy the script. Nobody — including the engine — can move the tokens alone. The engine proves the lock on chain and fails closed if the transaction is not `CONFIRMED` at the derived escrow address.
3. **The seller's own confirmation is the fiat signal.** There is no webhook for a WeChat/Alipay/bank transfer. `PAYMENT_CLAIMED` is only the buyer's claim (流水号 + optional receipt); tokens stay locked until the seller signs `confirm` after checking their own account — the UI repeats this as a red warning above the *Confirm received* button.
4. **PII never reaches the order book or the chain.** `GET /public/orders` carries only the trade terms. The seller's 收款 account (WeChat ID, Alipay account, or bank card + bank name) is revealed party-scoped by `POST /swaps/:id/payment-instructions`; only a per-swap `remark` code plus (optionally) a receipt `sha256` are stored on the swap event.
5. **Every mutating call is signed with the party's PQ (ML-DSA-87) `did:key`** — the same wallet key that signs chain transactions — over the sha256 of the canonical (sorted top-level keys) JSON body, plus a nonce + timestamp (±5 min): replay-protected, rate-limited to 10 signed requests per DID per minute (`sign.ts:17-19`). The demo's final legs (`release` / `complete`) are signed by the engine signer (`SETTLEMENT_ENGINE_DID`, an ML-DSA did:key ≈3.5 kB); nothing fires them automatically yet, so the capture script stands in for that operator job (§6.1).

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1. State machine — who advances each step

CNY rails (`wechat` / `alipay` / `bank` — seller-confirmed):

#	Status	Action (HTTP)	Who must sign	CNY-specific evidence
1	`MATCHED`	`POST /orders/:id/match`	buyer	`receiveAddress` (token payout address)
2	`ESCROW_LOCKED`	`POST /swaps/:id/transitions action=escrow_lock`	**seller**	`txHash` (escrow lock proof)
3	`PAYMENT_PENDING`	`POST /swaps/:id/payment-instructions`	buyer	`remark` (12 hex chars, per-swap)
4	`PAYMENT_CLAIMED`	`POST /swaps/:id/proof`	buyer	`txId` 流水号, optional `receipt` + `receiptSha256`
5	`PAYMENT_VERIFIED`	`POST /swaps/:id/confirm`	**seller**	the seller's own statement check
6	`ESCROW_RELEASED`	`POST /swaps/:id/transitions action=release`	**engine**	`txHash` (release proof)
7	`COMPLETED`	`POST /swaps/:id/transitions action=complete`	**engine**	— (seller already holds the CNY)

Freeze paths: either party may `POST /swaps/:id/dispute` while `PAYMENT_PENDING` / `PAYMENT_CLAIMED`; the admin then resolves with `outcome=release|refund`. `expire` / `refund` / `cancel` stay available before release so a failed deal can always return the tokens.

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2. The rails and how they are configured

Rail (`wantRail`)	Buyer pays	Seller verifies	Tokens released by
`wechat`	WeChat Pay to seller's WeChat ID	seller's own 流水 + remark	engine `release` + `complete`
`alipay`	Alipay to seller's account	seller's own 流水 + remark	engine `release` + `complete`
`bank`	bank transfer + remark	bank statement + remark	engine `release` + `complete`

Engine environment (`services/p2p-engine/src/server.ts`):

```
SETTLEMENT_CNY_RAILS=wechat,alipay,bank  # unset → all three; "" → rail off
SETTLEMENT_CNY_REMARK_TTL=900            # seconds → payBy deadline in instructions
SETTLEMENT_ENGINE_DID=did:key:z446Gmp1… # ML-DSA-87 did:key (≈3.5 kB); only signer of release/complete
SETTLEMENT_ENGINE_PUBKEY=050101010a20…  # prefixed ML-DSA pub (hex) → 2-of-3 escrow address derivation
SETTLEMENT_STORE=mem|pg                  # mem for demos, pg for production
SETTLEMENT_ADMIN_TOKEN=adm               # /orders, /swaps, dispute/resolve
CORS_ORIGIN=http://localhost:18081       # wallet web app origin
```

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3. Running the demo yourself

```
# 1. workspace packages the engine imports
for pkg in chain-discovery did p2p-protocol record-sig; do ( cd packages/$pkg && npm run build ); done

# 2. engine bundle (esbuild → dist/server.bundle.mjs)
( cd services/p2p-engine && npm run build && npm run bundle )

# 3. wallet web app with the engine URL inlined by `expo export`
( cd expo-app && EXPO_PUBLIC_P2P_ENGINE_URL=http://localhost:18089 npm run web:build )
echo http://localhost:18089 > e2e/web-build/.p2p-url

# 4. CNY legs: 14 screenshots per rail → assets/screenshots/p2p-cny-{wechat,alipay,bank}-NN-*-en.png
( cd e2e && RAILS=wechat,alipay,bank node capture-p2p-cny.mjs )

# 5. render the PDF
node docs/p2p-demo/scripts/gen-p2p-all-pdf.mjs
```

Or all of it through the one entry point: `./p2pdemo.sh capture walkthrough` (see `p2pdemo.sh --help`). The captures are self-contained: no L0/L1 — a CNY swap has no chain-side dependency beyond the escrow lock proof the engine skips when `deps.chain` is unset. `RAILS=wechat` (or `alipay` / `bank`) limits the run to one rail.

Run facts used in the screenshots below:


```
{ "wechat": { "swapId": "swap-a13efab115858720", "txId": "42034516068EAEC2", "remark": "f71ade2d23e9"
},
  "alipay": { "swapId": "swap-f8226be10089ddad", "txId": "200E31282969161F6B32A2CF1C93", "remark":
"91b6eeced27f" },
  "bank":   { "swapId": "swap-ad4639ba7976dd97", "txId": "30291CF907571417", "remark": "20e892c372ce"
},
  "capturedAt": "2026-10-12T10:45:52Z" }
```

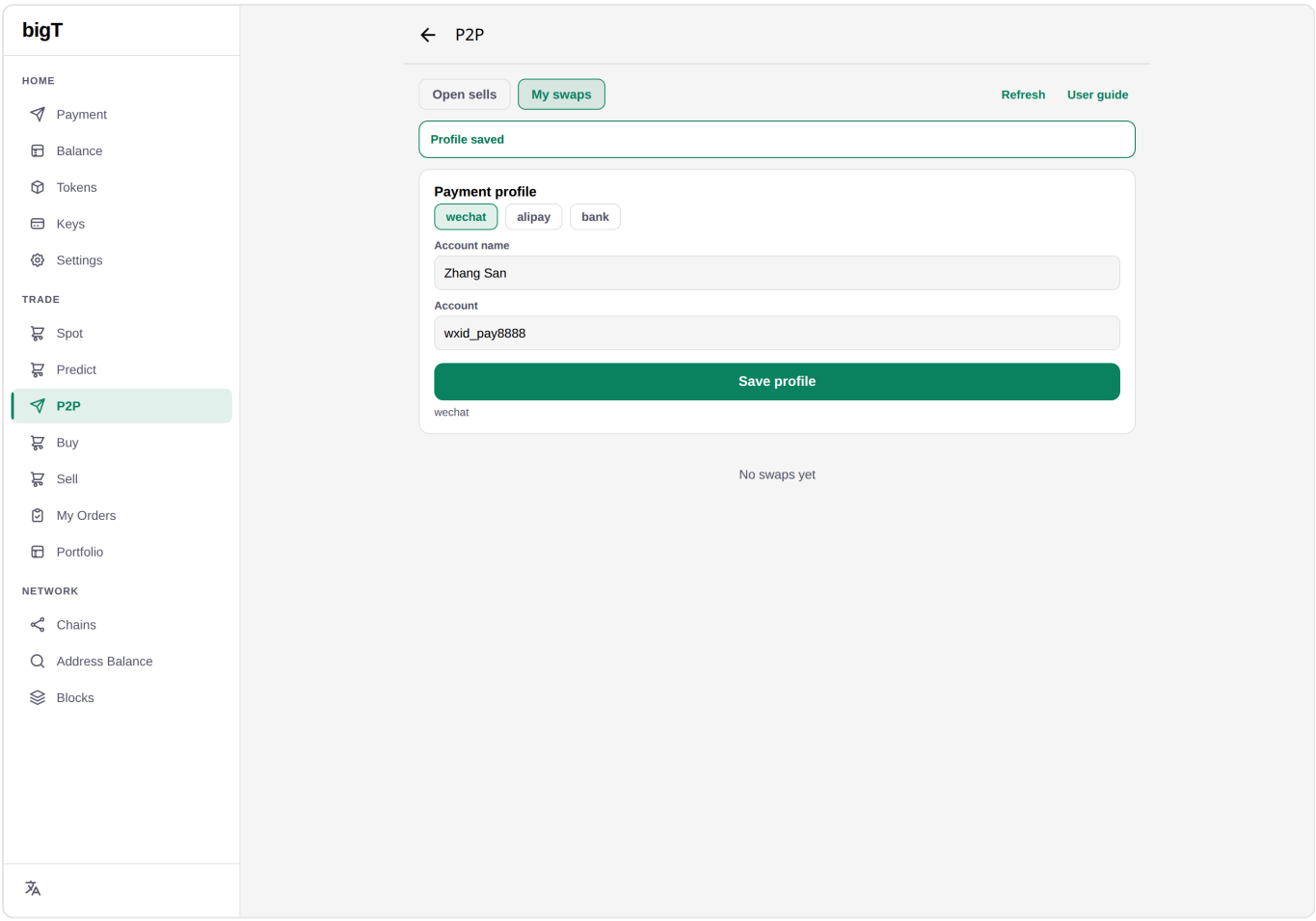

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4. Walkthrough — WeChat Pay (CNY)

The CNY rails need one extra step at the start (the seller's collection profile) and replace the webhook leg with `instructions → proof → confirm`. The seller is paid **first** (the buyer transfers CNY directly to the seller's own WeChat account); the tokens are released only after the seller confirmed.

Step 1 — Seller saves a collection profile

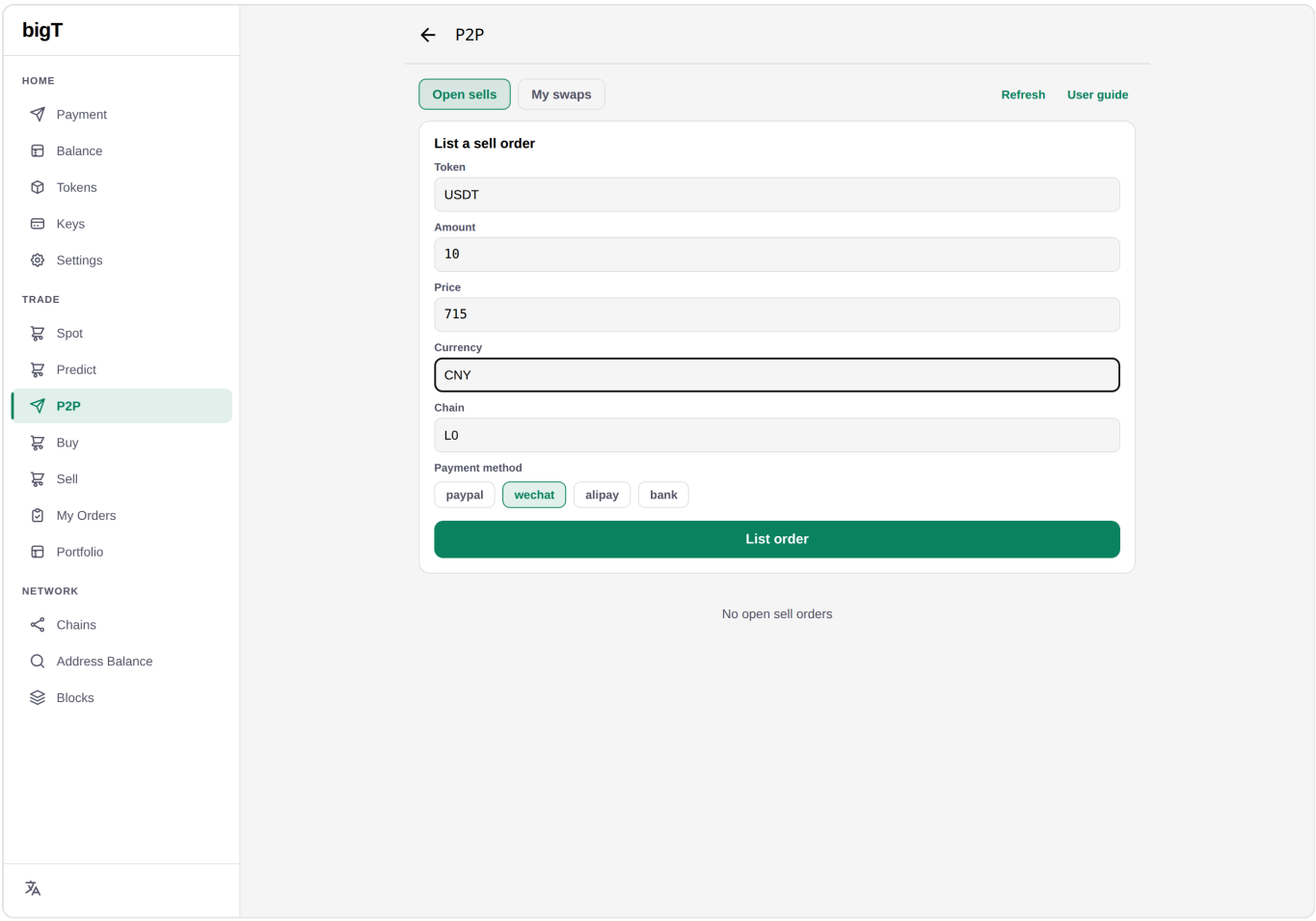
The seller opens **P2P → My swaps**, picks the rail (`wechat`), and enters the 收款 name and account (WeChat ID). The profile is stored server-side under the seller's DID and is **not** part of the public order book — it is only revealed to the matched buyer by the instructions call.



Step 2 — Seller lists 10 USDT for 715 CNY on the wechat rail

Open sells → List a sell order: token `USDT`, amount `10`, price `715`, currency `CNY`, chain `L0`, payment method **wechat** (selecting a CNY rail switches the currency to CNY). The signed `POST /orders` payload carries no PII:

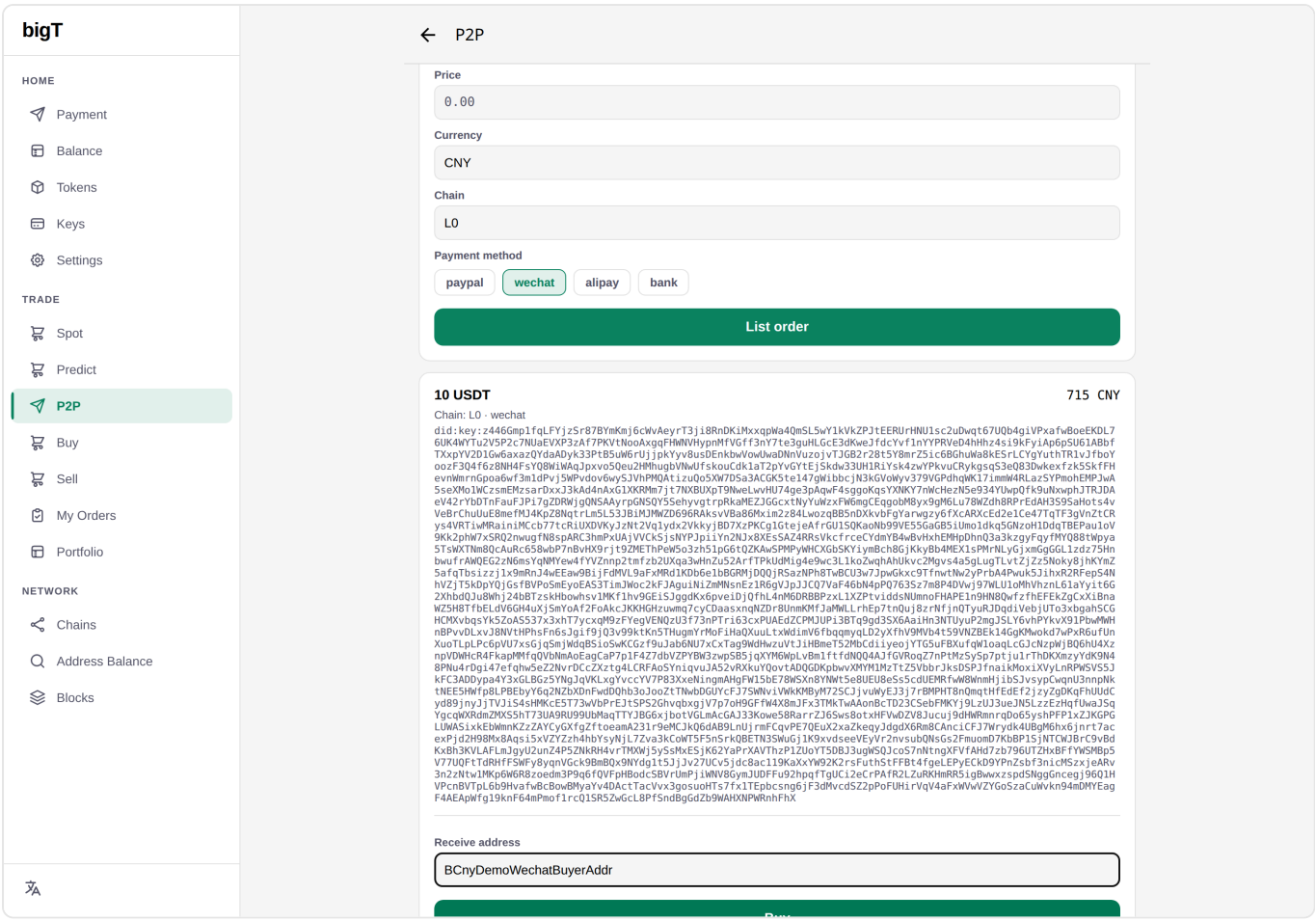
```
{ "sellerDid": "did:key:z446G…(ML-DSA did, ≈3.5 kB)", "giveToken": "USDT",
  "giveAmount": "10", "giveChain": "L0", "wantCurrency": "CNY",
  "wantAmount": "715", "wantRail": "wechat", "validUntil": 1791510000,
  "nonce": "…16+ chars…", "timestamp": 1791509100000,
  "signature": "…hex SignatureBundle over sha256(canonical JSON)…"}
```



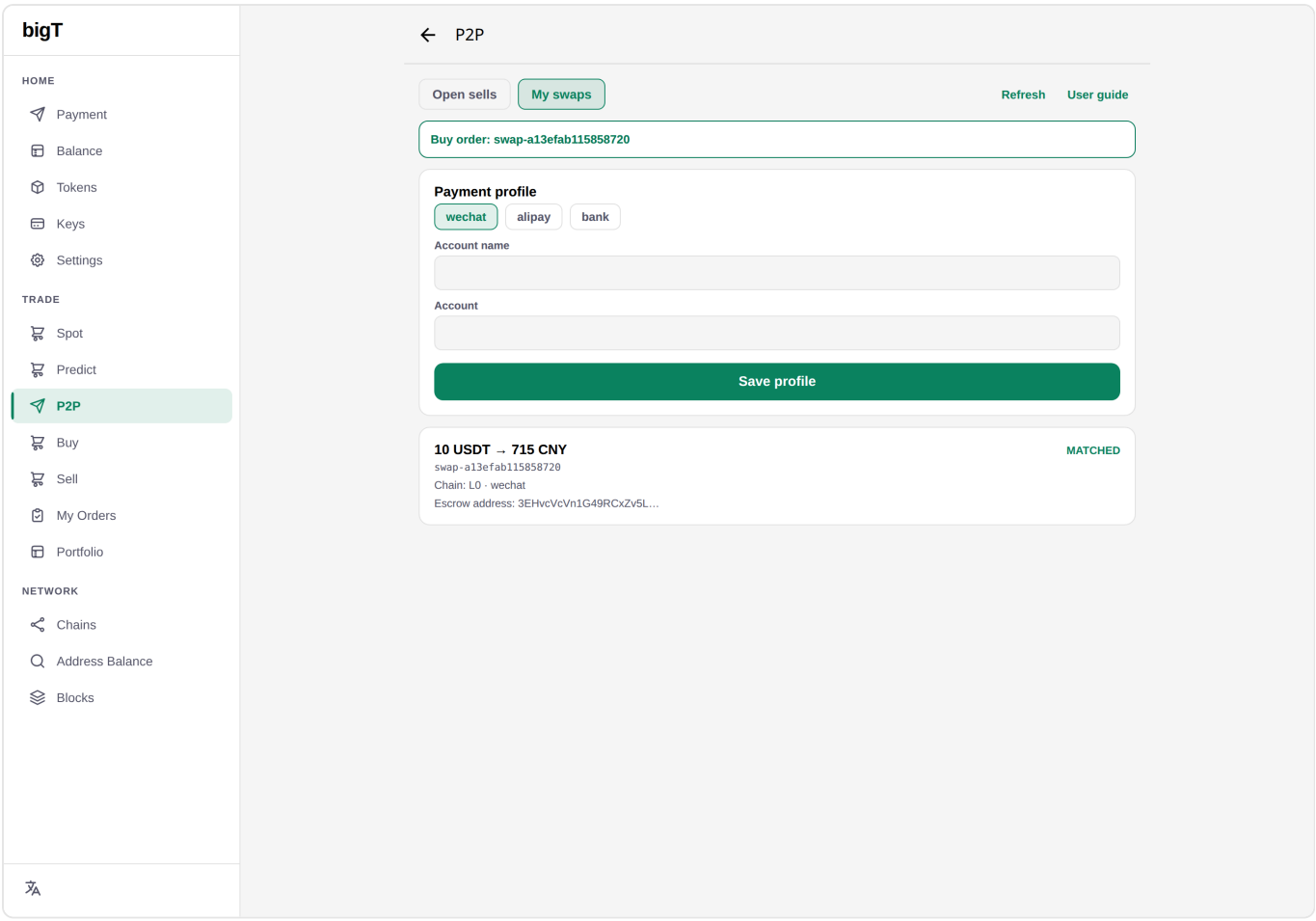
Step 3 — Buyer matches the order

The buyer (fresh wallet, no profile needed) sees the order in **Open sells**

and opens the buy panel. On a CNY order only a receive address is required:

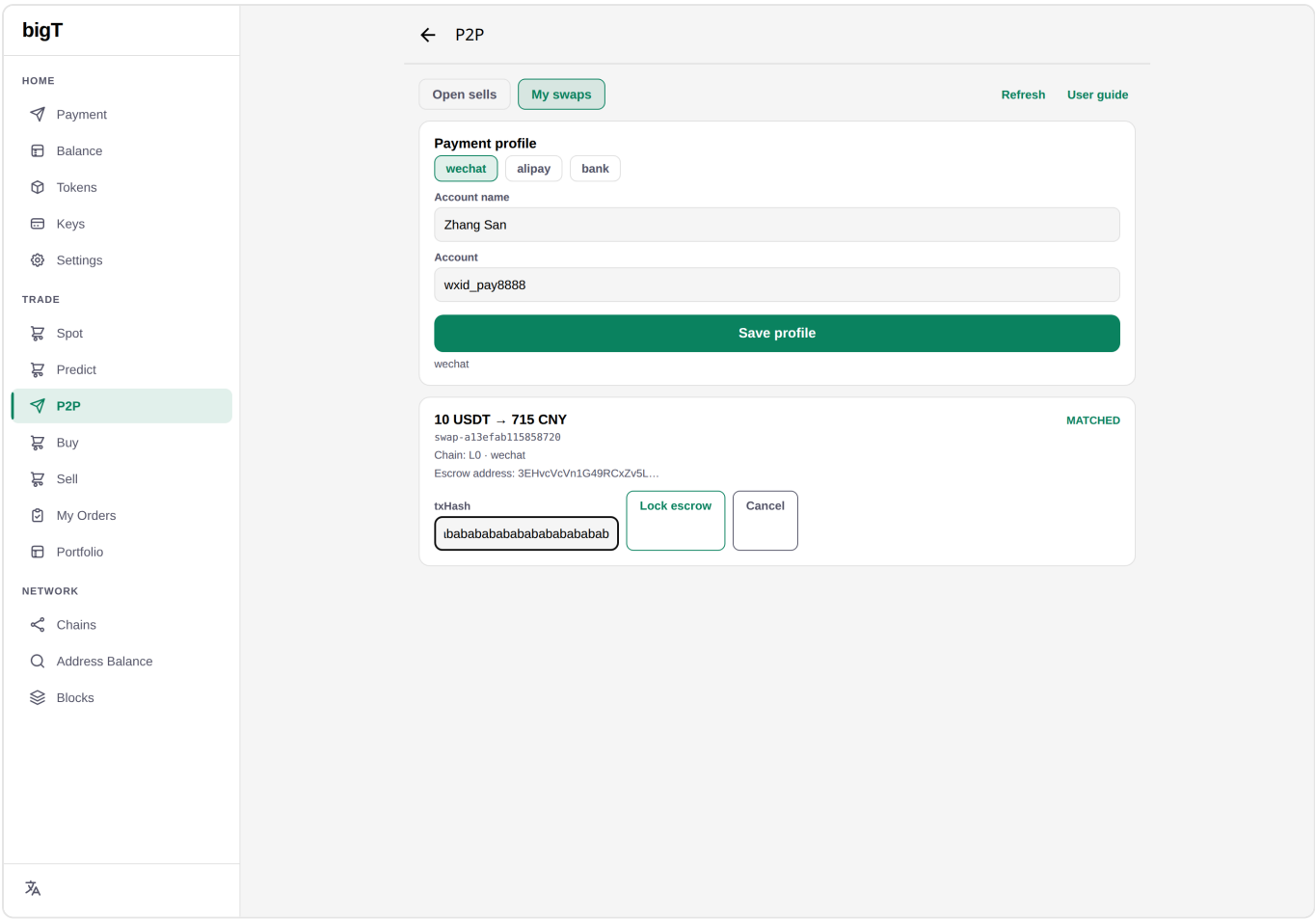


After **Confirm buy** the engine mints the swap at `MATCHED` and derives the 2-of-3 escrow address from the seller's and buyer's PQ dids plus the engine's prefixed ML-DSA public key (`SETTLEMENT_ENGINE_PUBKEY`):

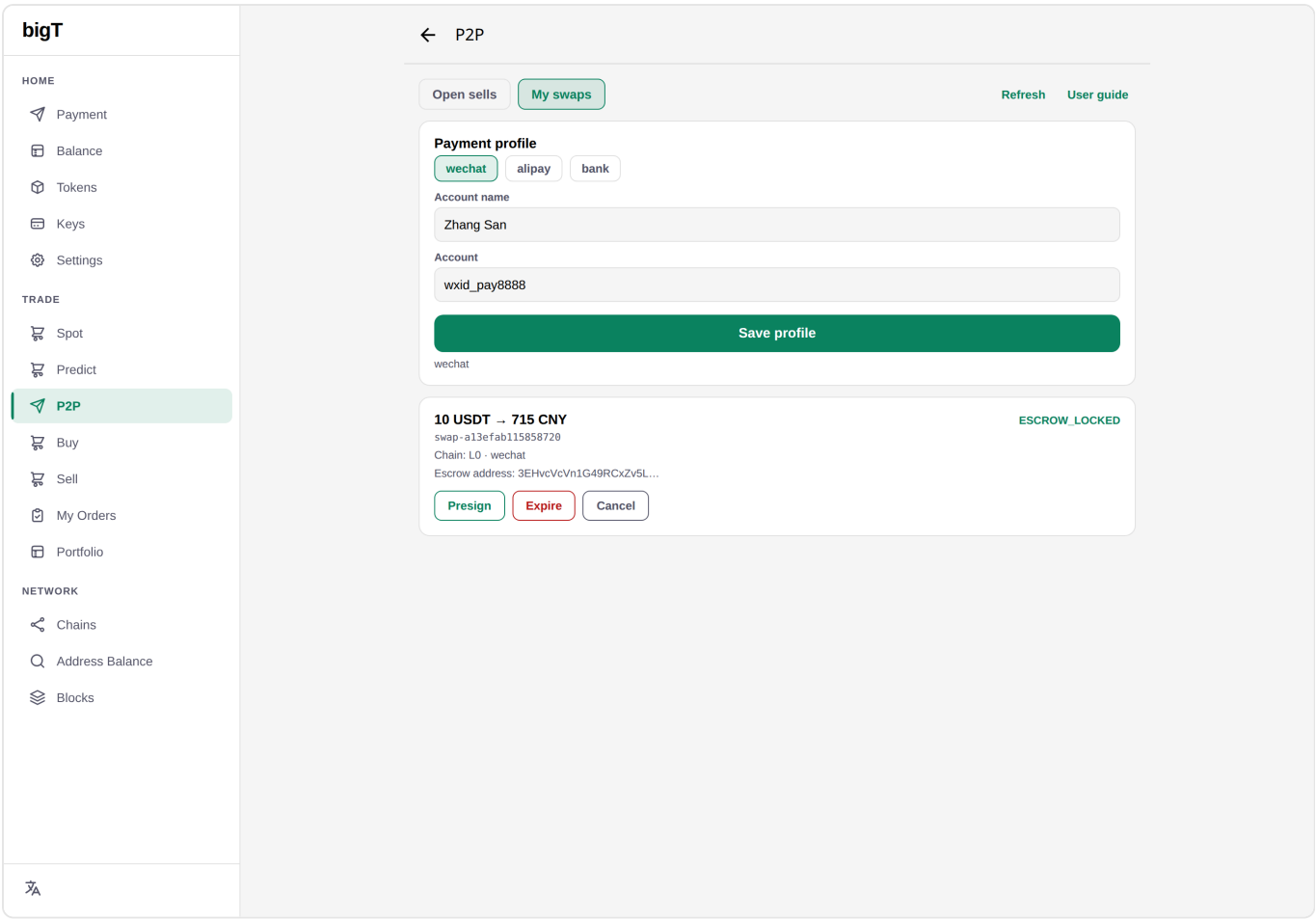


Step 4 — Seller locks the token escrow

Back on the seller's **My swaps** tab, the seller pastes the escrow funding transaction hash and presses **Lock escrow** — a seller-signed `action=escrow_lock` transition. With `deps.chain` configured the engine verifies the lock on chain; in the demo (`SETTLEMENT_STORE=mem`, no chain) it is recorded as-is.



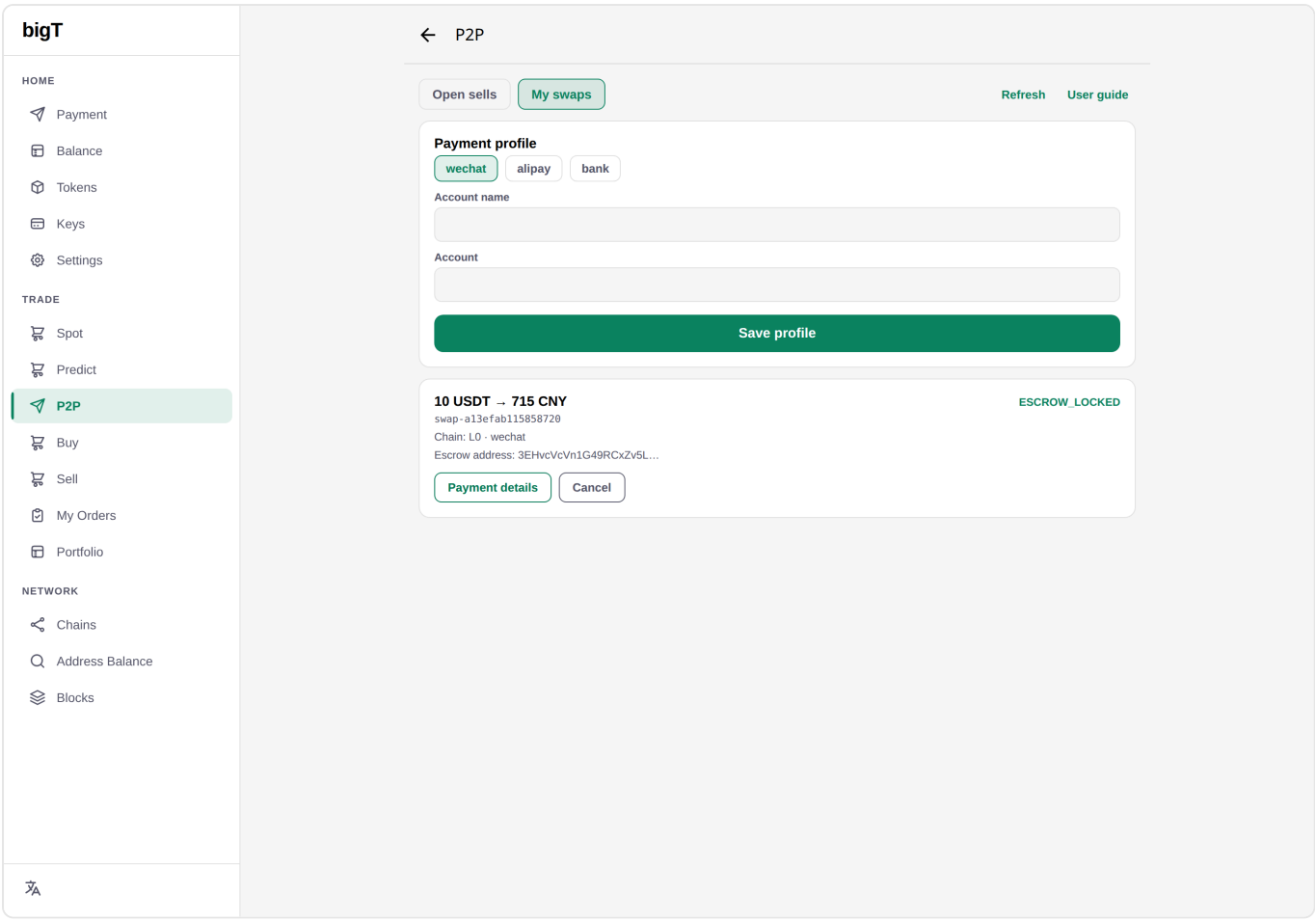
Status advances to `ESCROW_LOCKED`. **Nothing has been paid yet** — this is the point of no return for the seller's tokens.



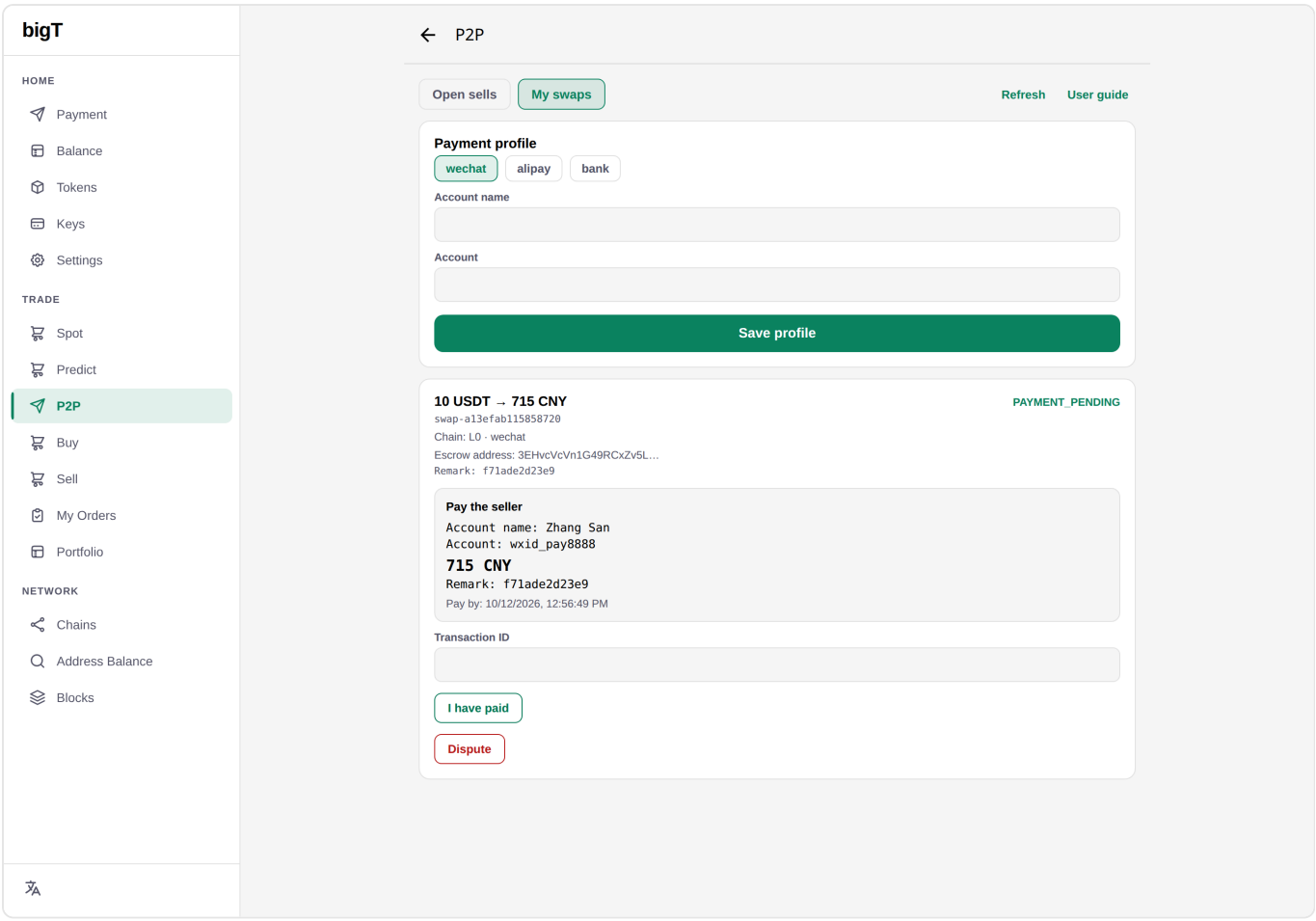
Step 5 — Buyer pulls the signed payment instructions

The buyer's card now shows a **Payment details** button. Pressing it issues the buyer-signed `POST /swaps/:id/payment-instructions`, which is idempotent while `PAYMENT_PENDING` and reveals the seller's profile + a fresh per-swap `remark` the transfer *must* carry:

```
{ "method": "wechat", "rail": "wechat", "accountName": "Zhang San",
  "account": "wxid_pay8888", "amount": "715", "currency": "CNY",
  "remark": "f71ade2d23e9", "payBy": 1791510060 }
```

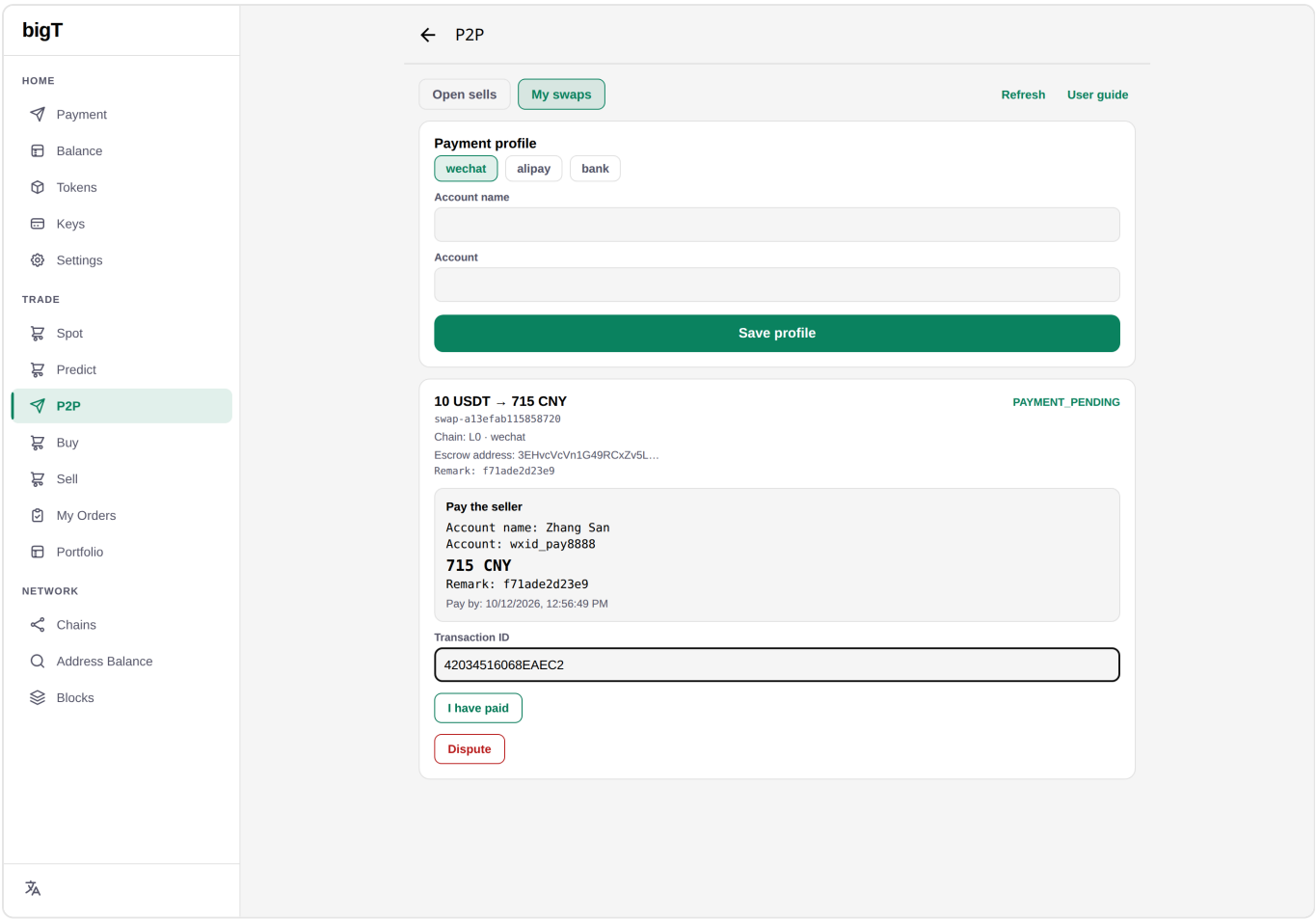


The instructions render **inline in the swap card** (amount, account, remark, deadline) — plus an optional QR payload when the profile stores one:

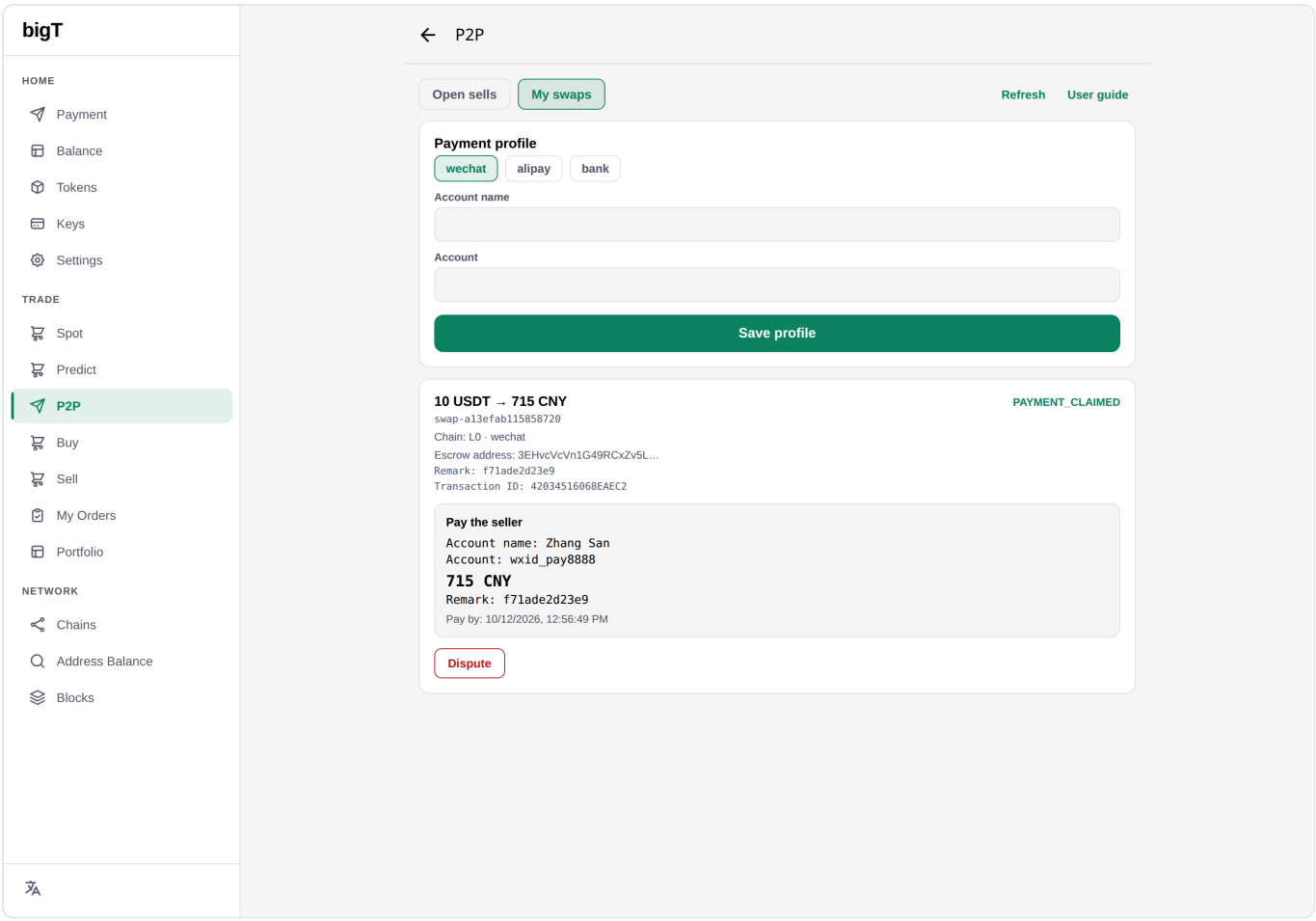


Step 6 — Buyer pays in WeChat and claims with the 流水号

The buyer transfers `715 CNY` in their own WeChat app, pastes the transaction number (流水号, starts `4203`) into **Transaction ID**, and presses **I have paid** (`POST /swaps/:id/proof`). The optional receipt image is stored as-is in the proof store; only its `sha256` is anchored on the swap event.



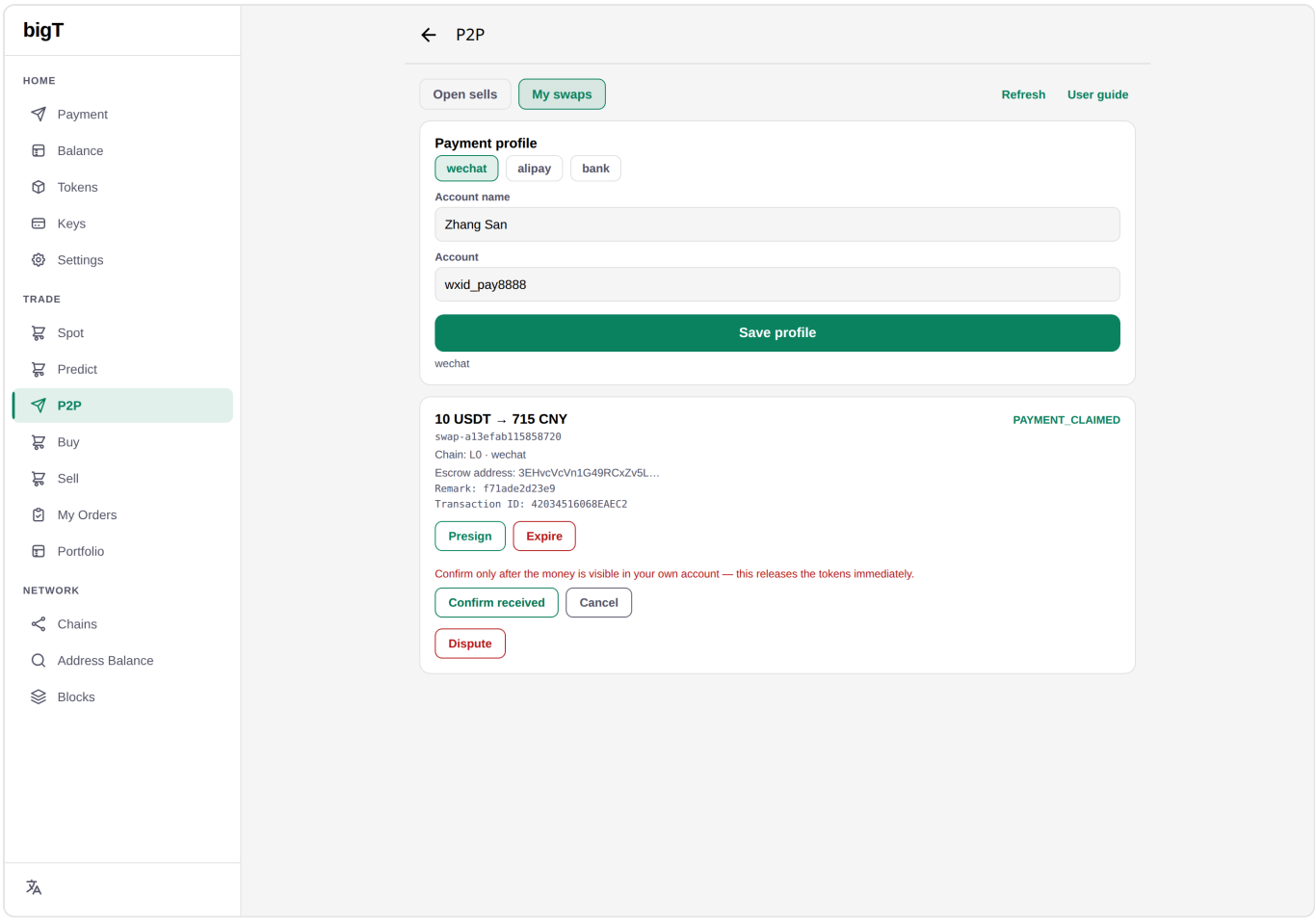
Status is now `PAYMENT_CLAIMED` — **the tokens are still locked.** This state exists purely as the buyer's claim.



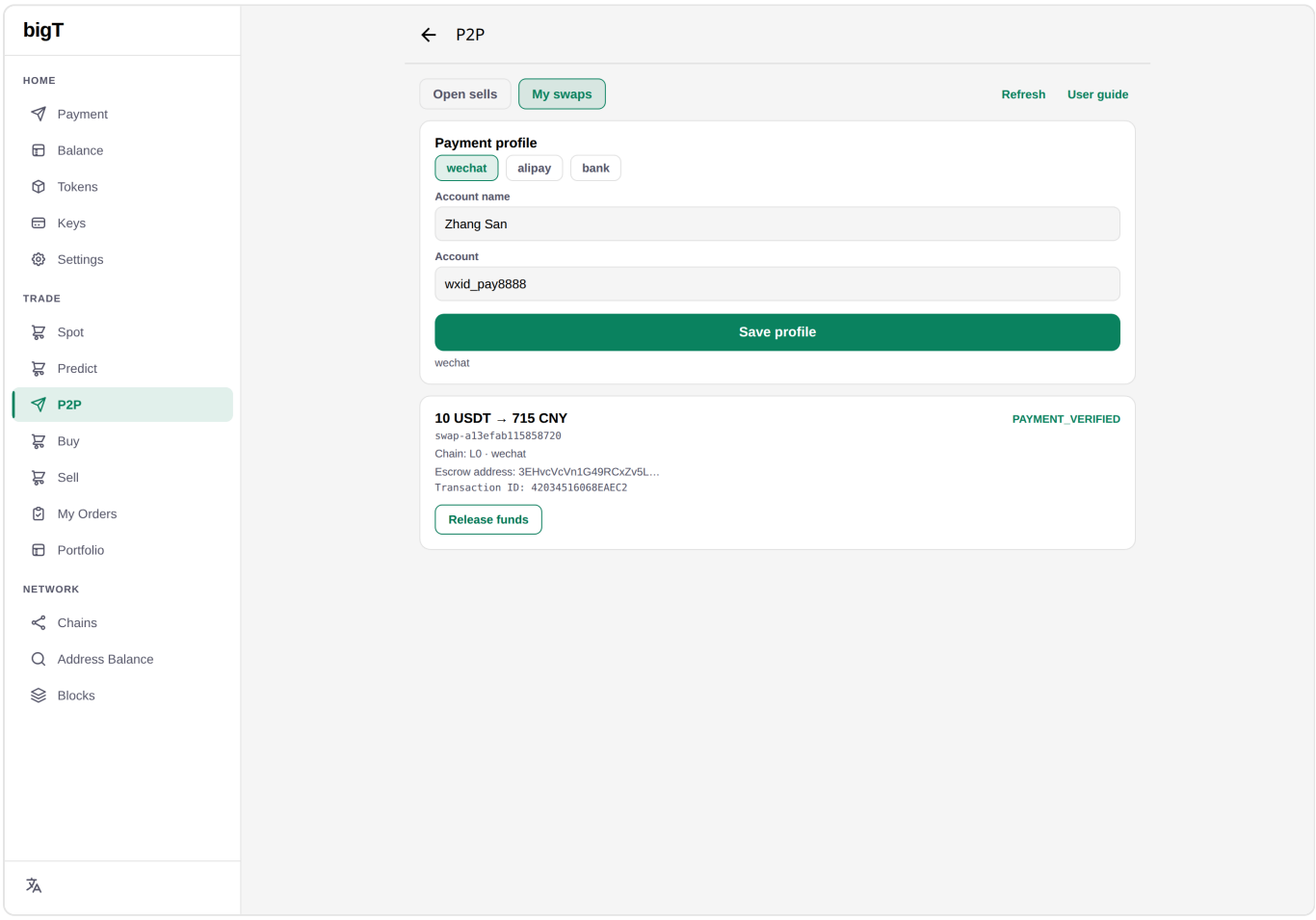
Step 7 — Seller verifies on their own statement, then confirms

The seller's card shows the buyer's `流水号` and the remark. The seller checks **amount + remark + time + 流水号 in their own WeChat account** (screenshots never count), and only then presses **Confirm received** — a seller-signed `POST /swaps/:id/confirm`. The red warning above the button is the whole safety story of the CNY rail:

> Confirm only after the money is visible in your own account — this releases the tokens immediately.



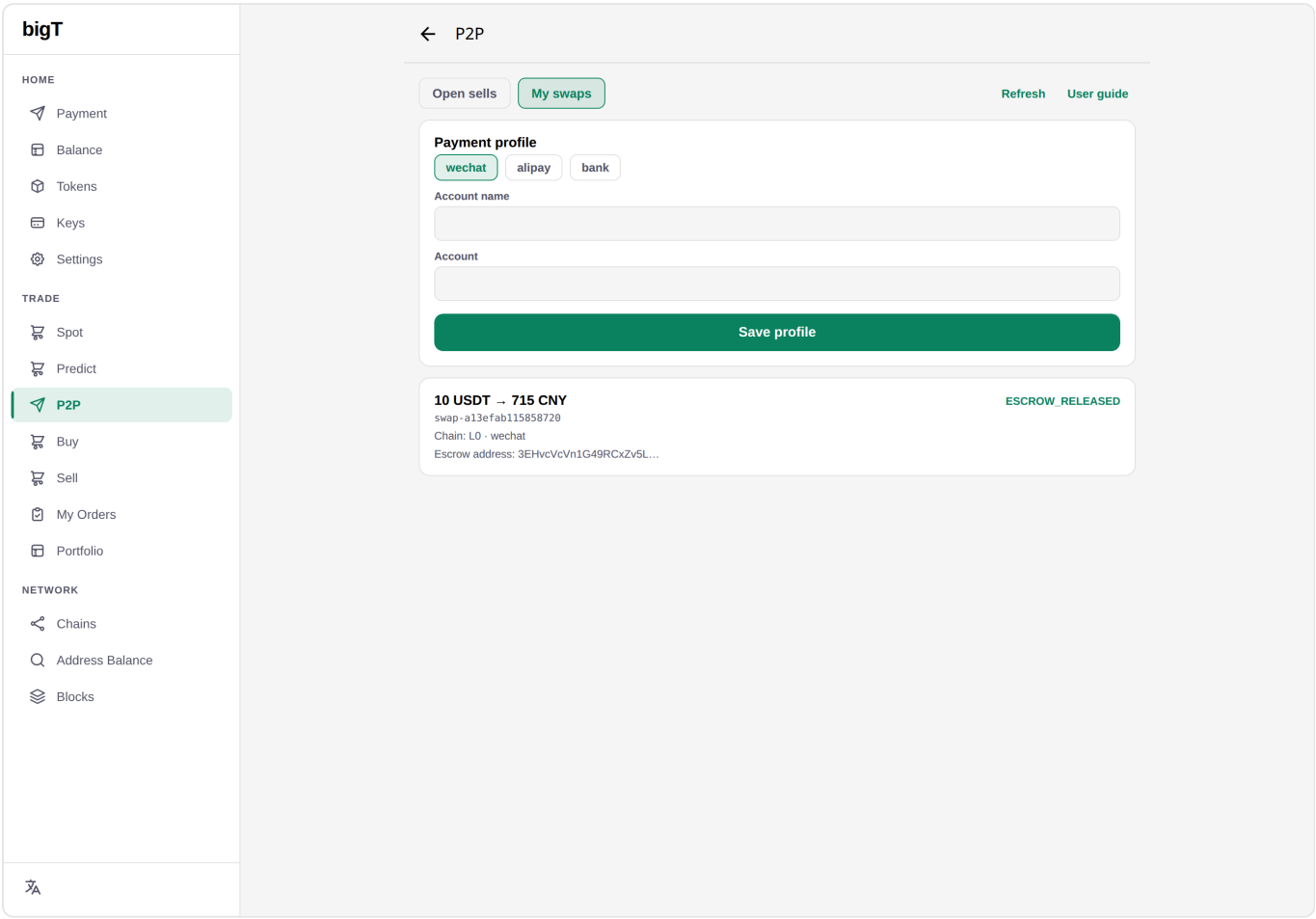
`PAYMENT_VERIFIED`:

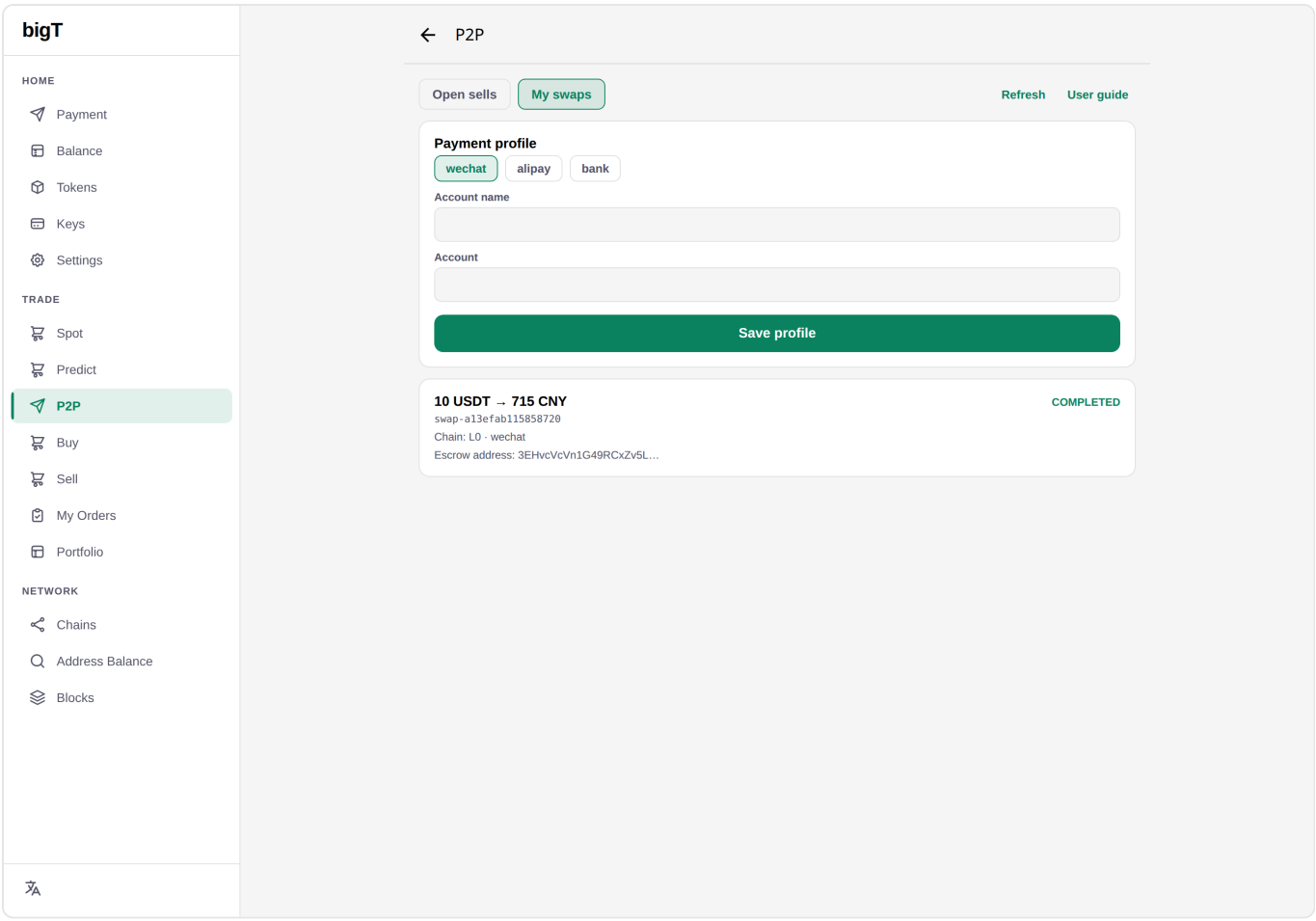


Step 8 — Engine releases the tokens, swap completes

`release` and `complete` are engine-signed actions — the wallet deliberately holds no key for them (§6.1). The capture script stands in for the operator / release job and signs both with the `SETTLEMENT_ENGINE_DID` key:

```
POST /swaps/swap-…/transitions   # did = engine, action=release, txHash=…
→ { "status": "ESCROW_RELEASED" }
POST /swaps/swap-…/transitions   # did = engine, action=complete
→ { "status": "COMPLETED" }
```

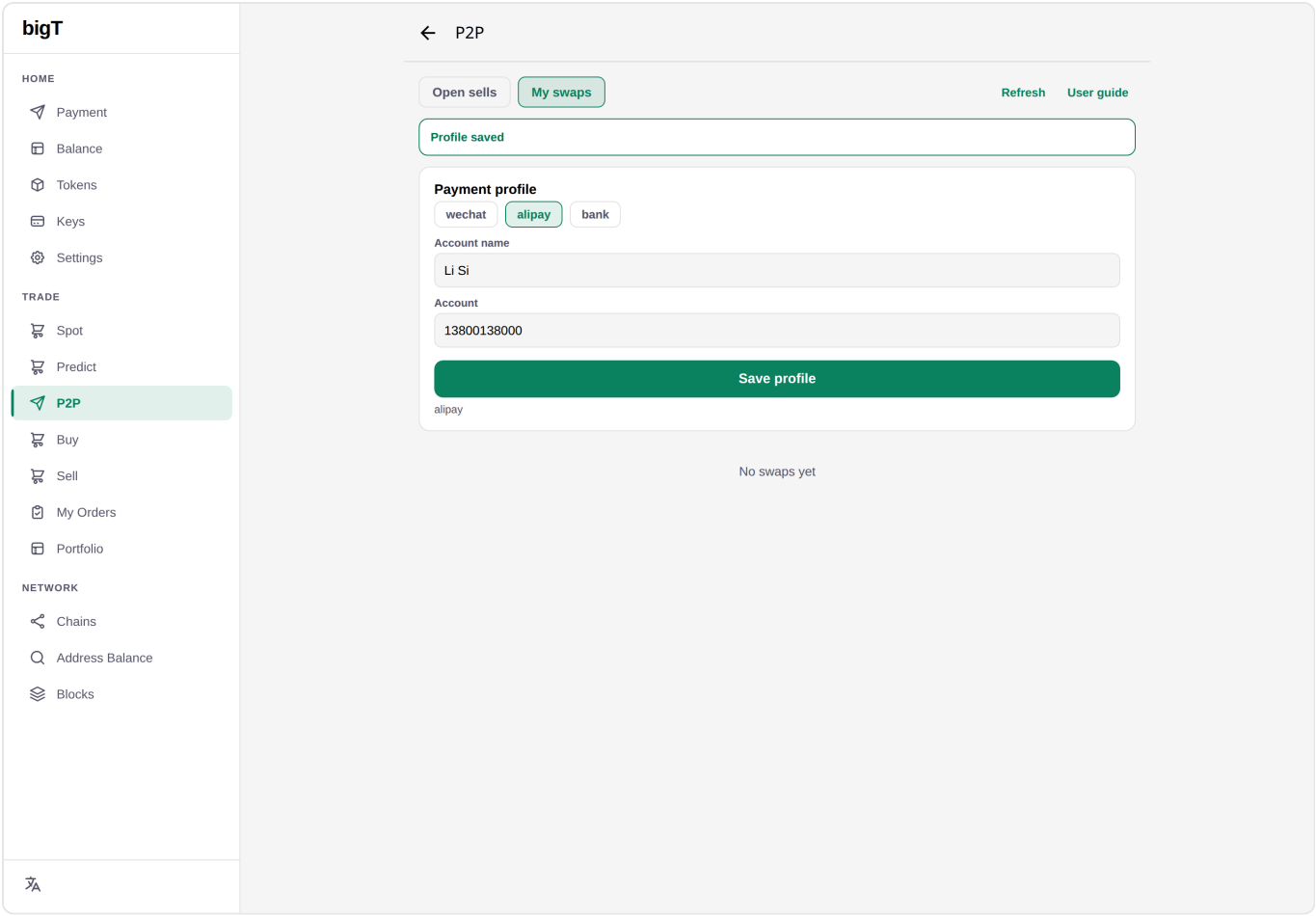




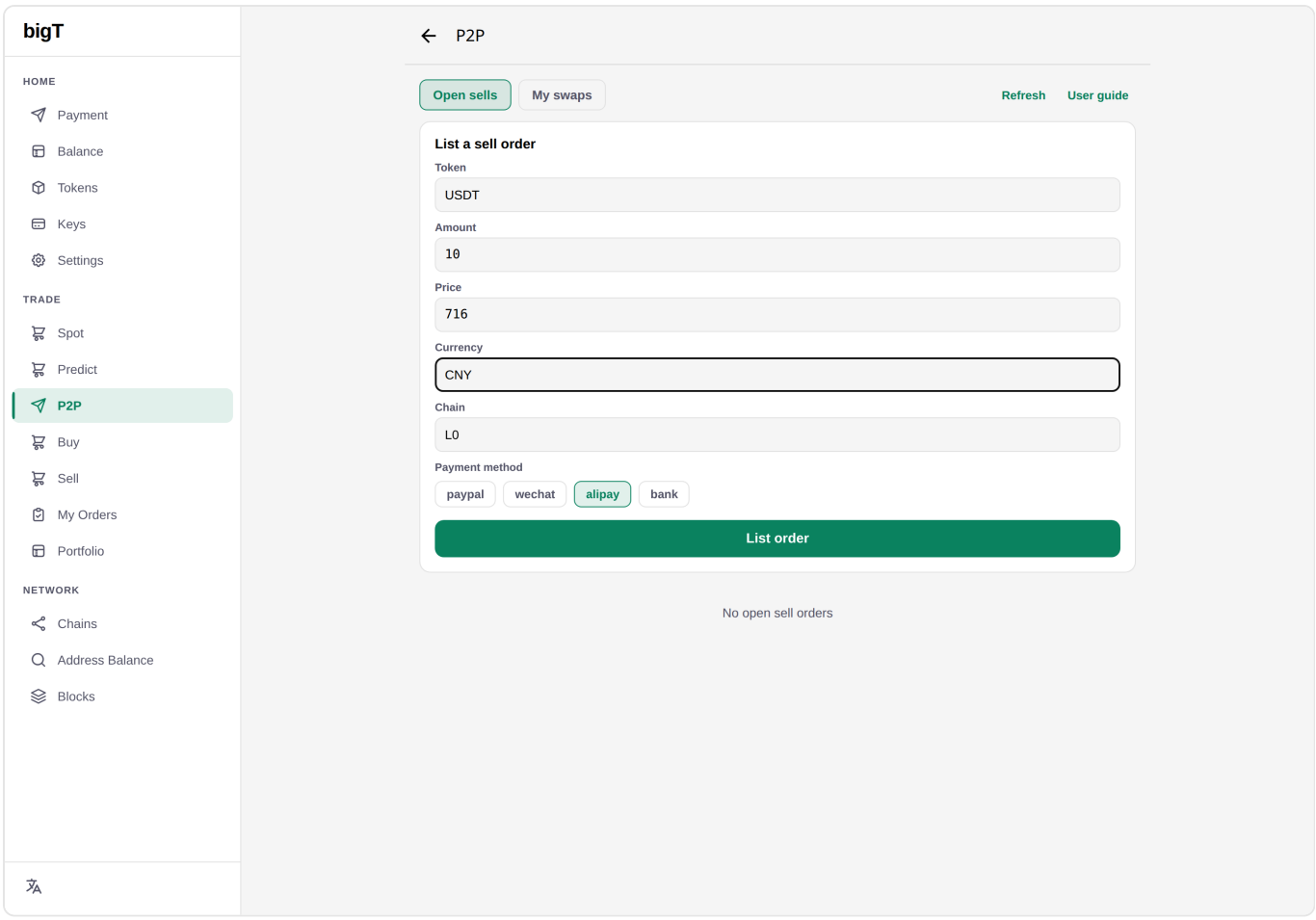
5. Walkthrough — Alipay (same shape, different rail)

The Alipay leg is the **same state machine** with `wantRail=alipay`: the seller saves an Alipay collection profile (account = 手机号 / Alipay logon), lists USDT for CNY on the alipay chip, and the buyer pays in their own Alipay app. The 流水号 is Alipay's 28-digit trade number (starts `20`). Fresh wallets are used so the screenshots stand alone.

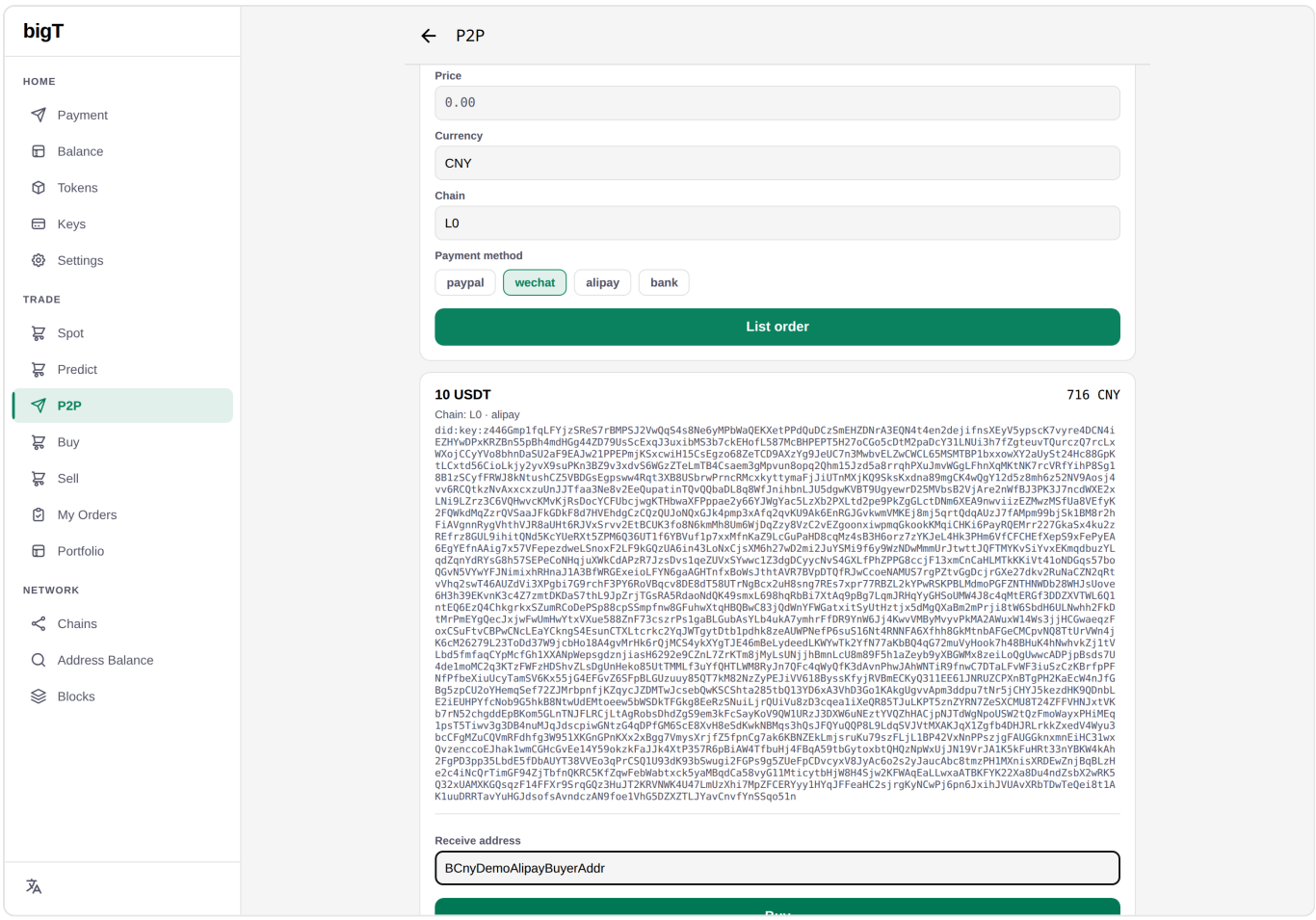
Step 1 — Seller saves the Alipay profile



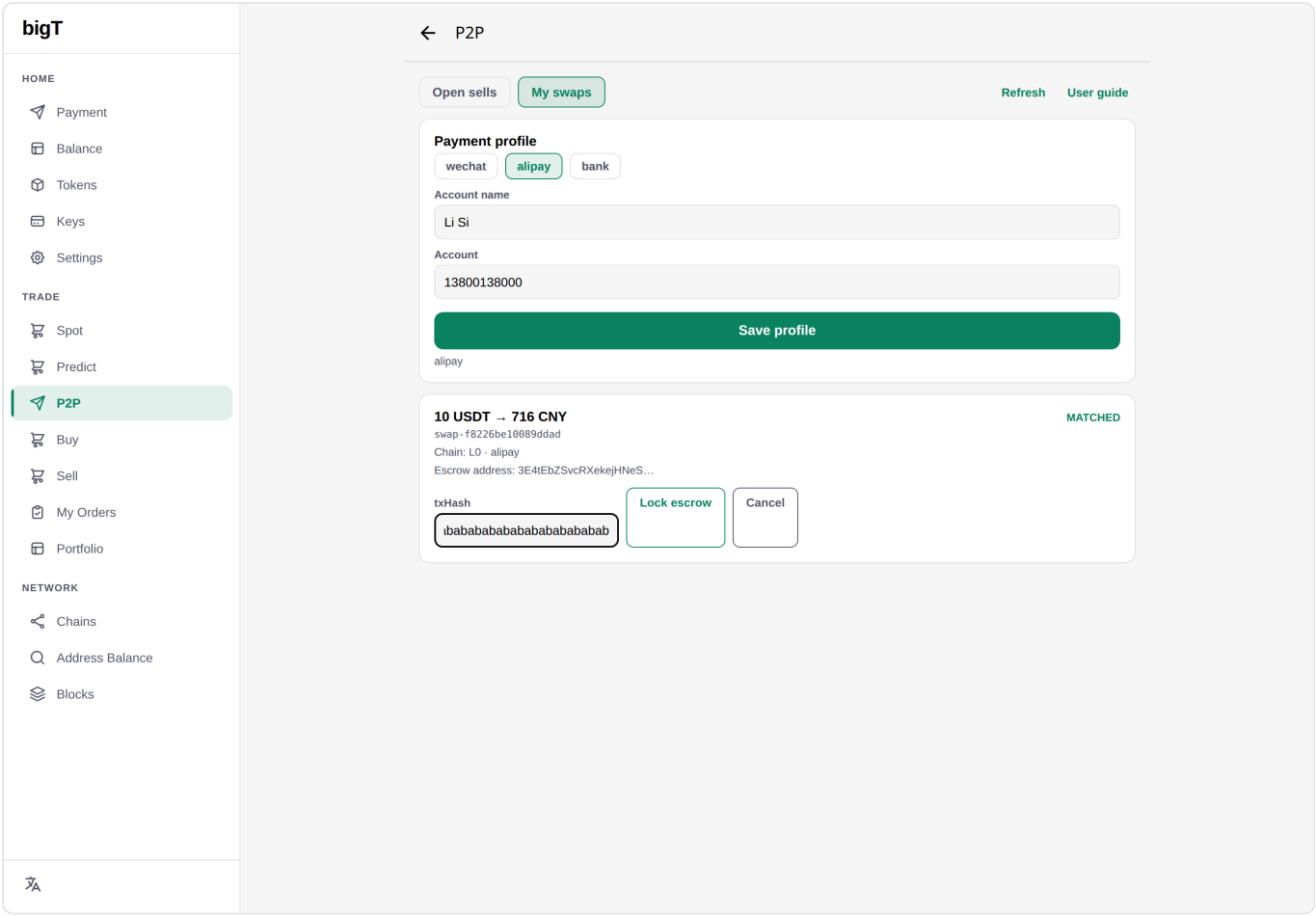
Step 2 — Seller lists 10 USDT for 716 CNY on the alipay rail

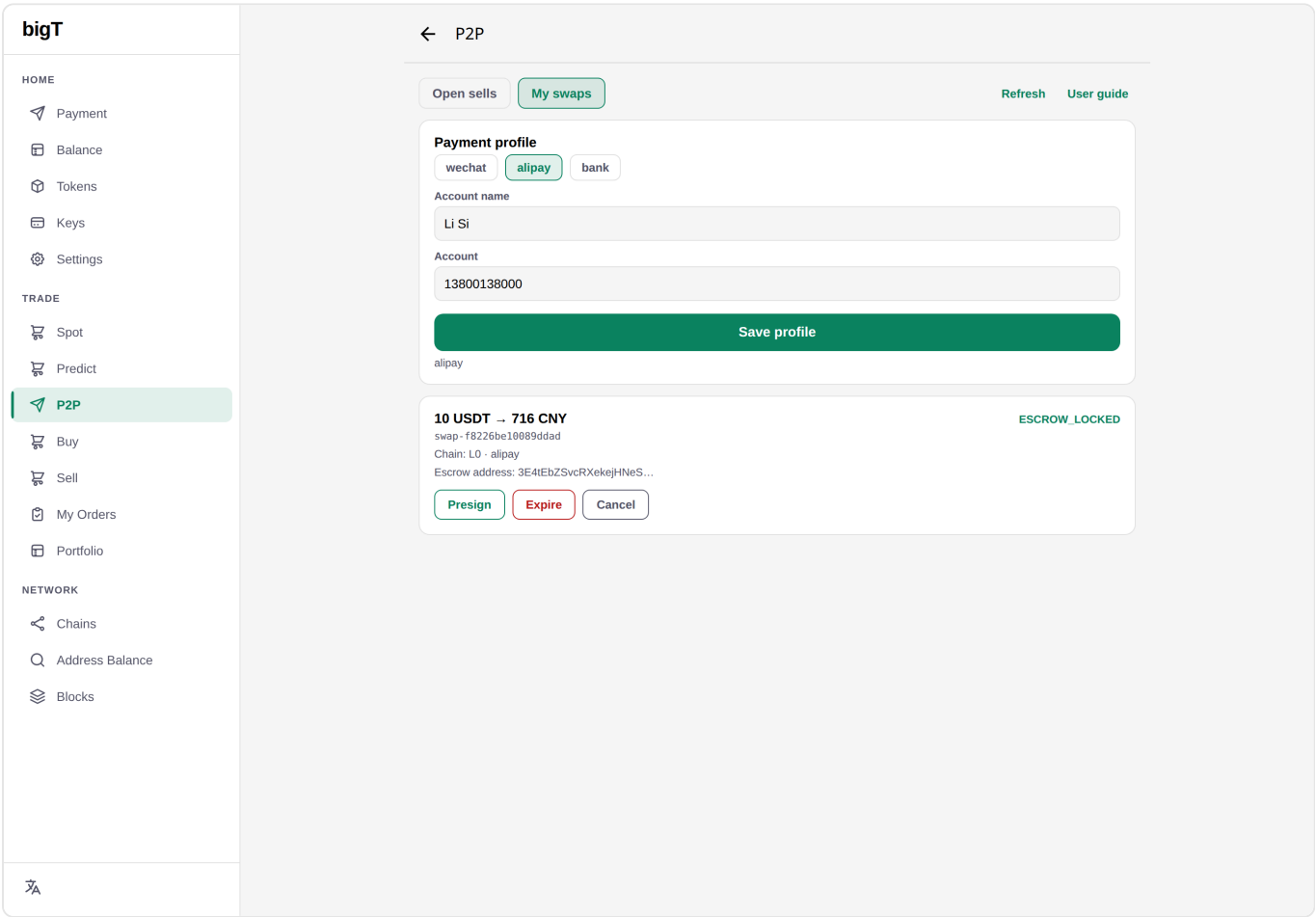


Step 3 — Buyer matches



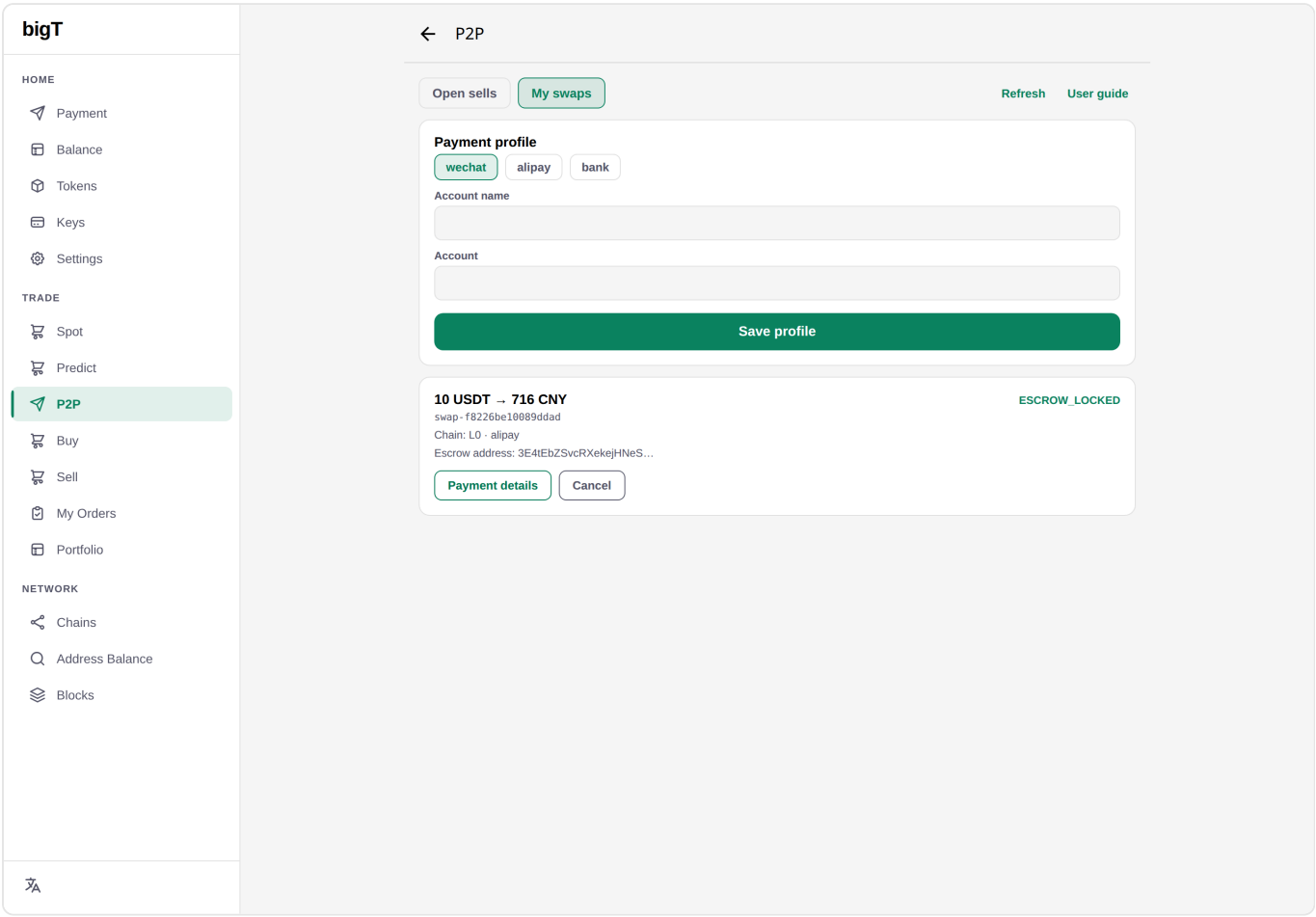
Step 4 — Seller locks the escrow

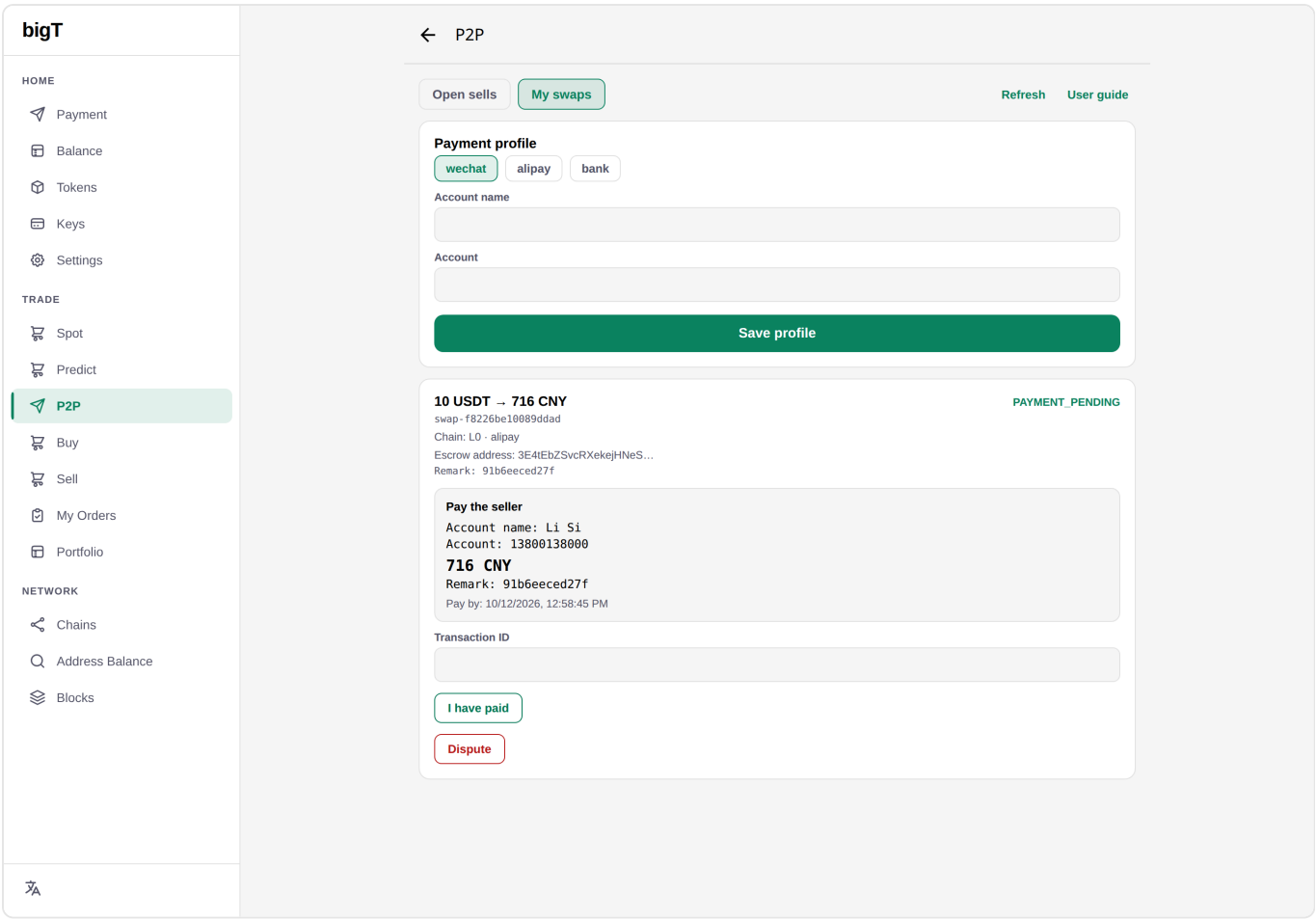




Step 5 — Buyer pulls the Alipay payment instructions

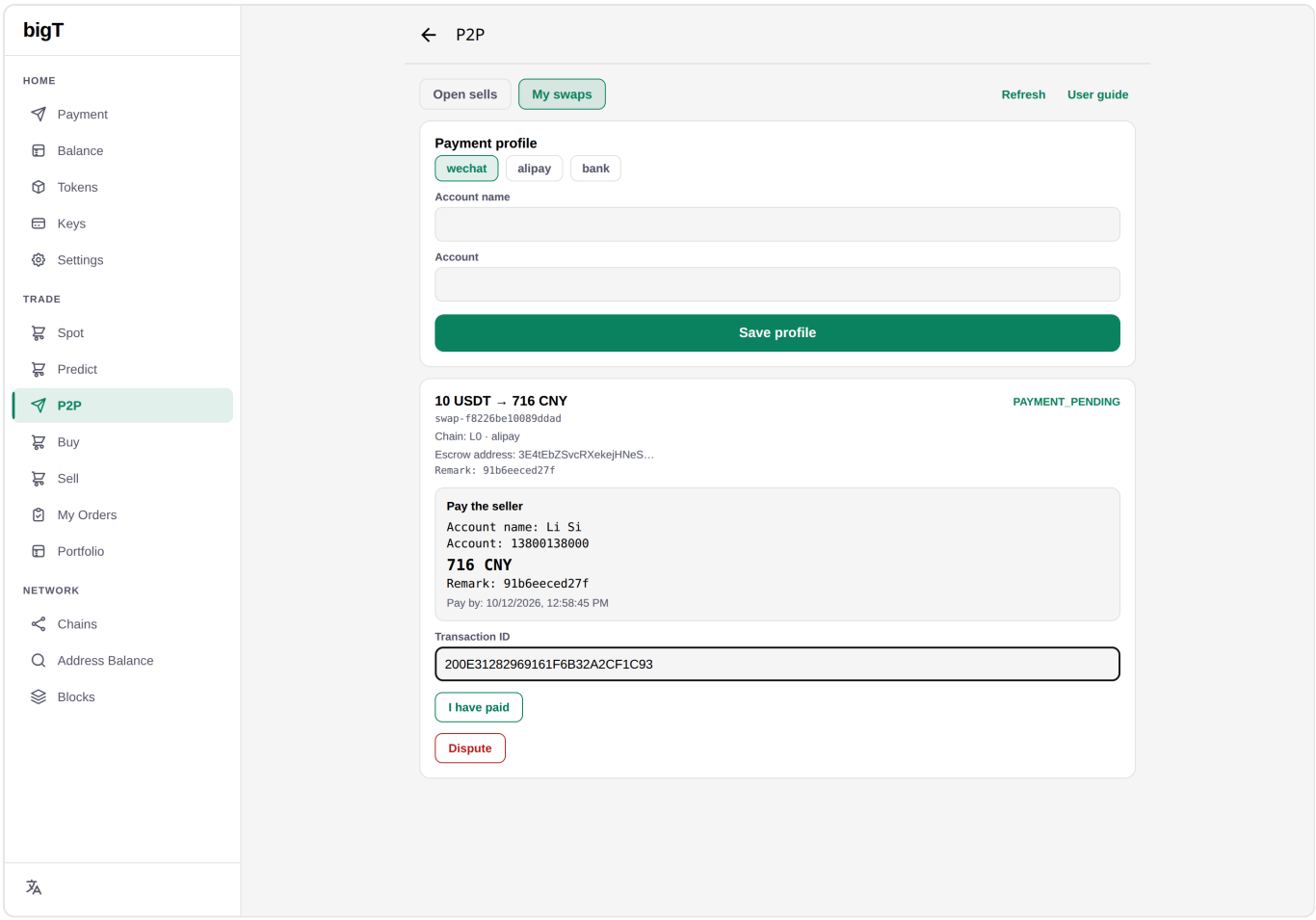

```
{ "method": "alipay", "rail": "alipay", "accountName": "Li Si",
  "account": "13800138000", "amount": "716", "currency": "CNY",
  "remark": "91b6eeced27f", "payBy": … }
```

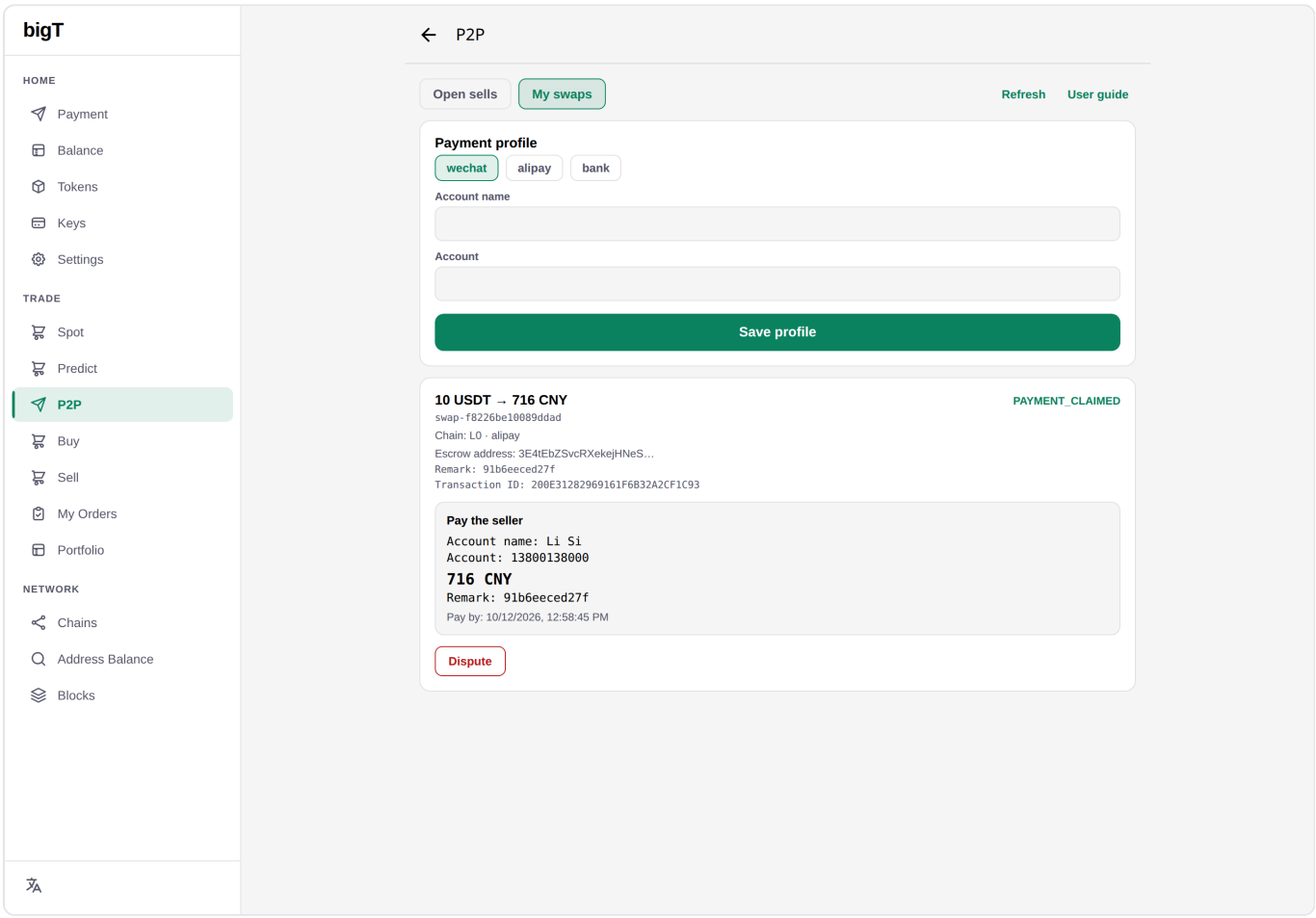





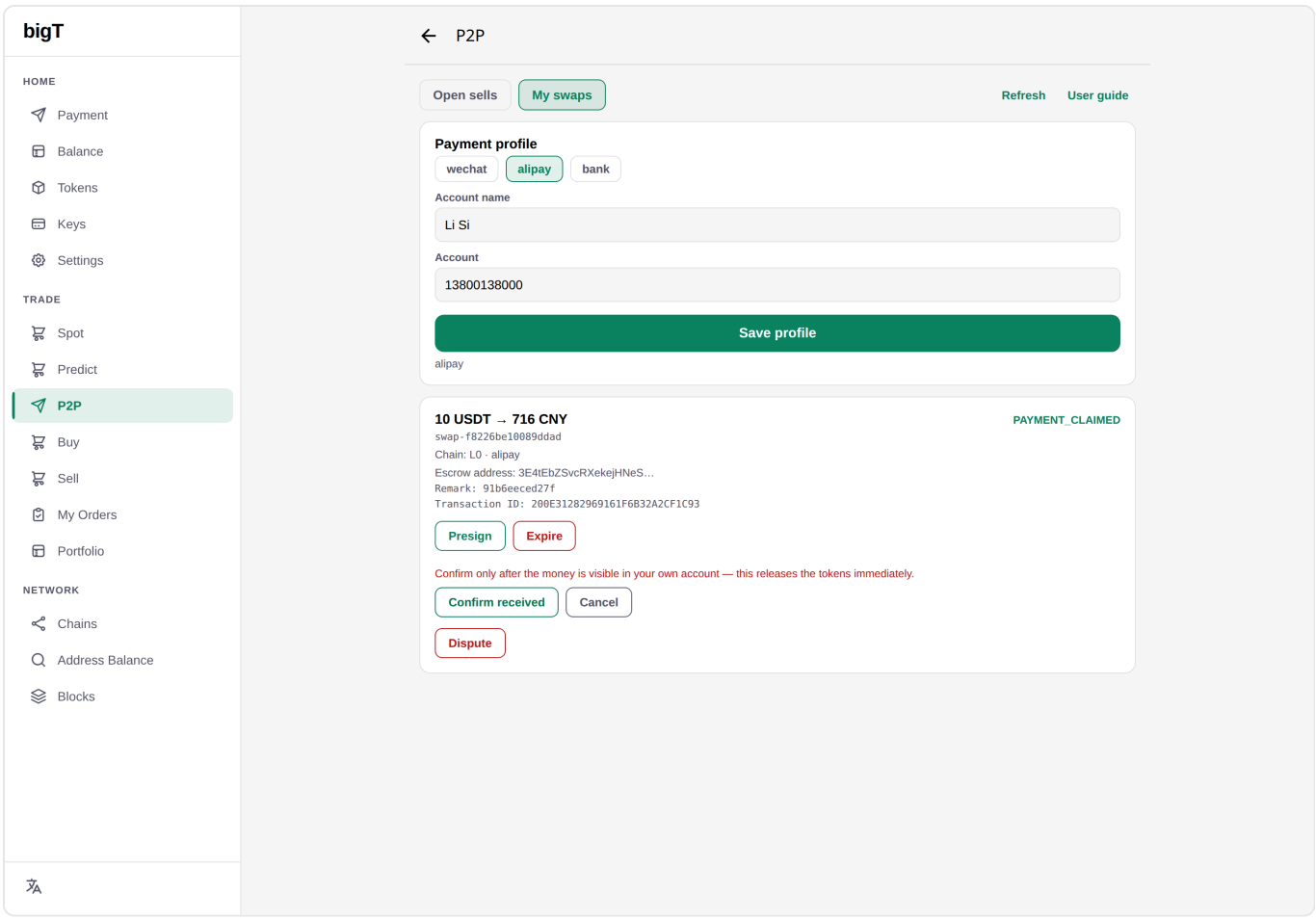
Step 6 — Buyer pays in Alipay and claims with the trade number

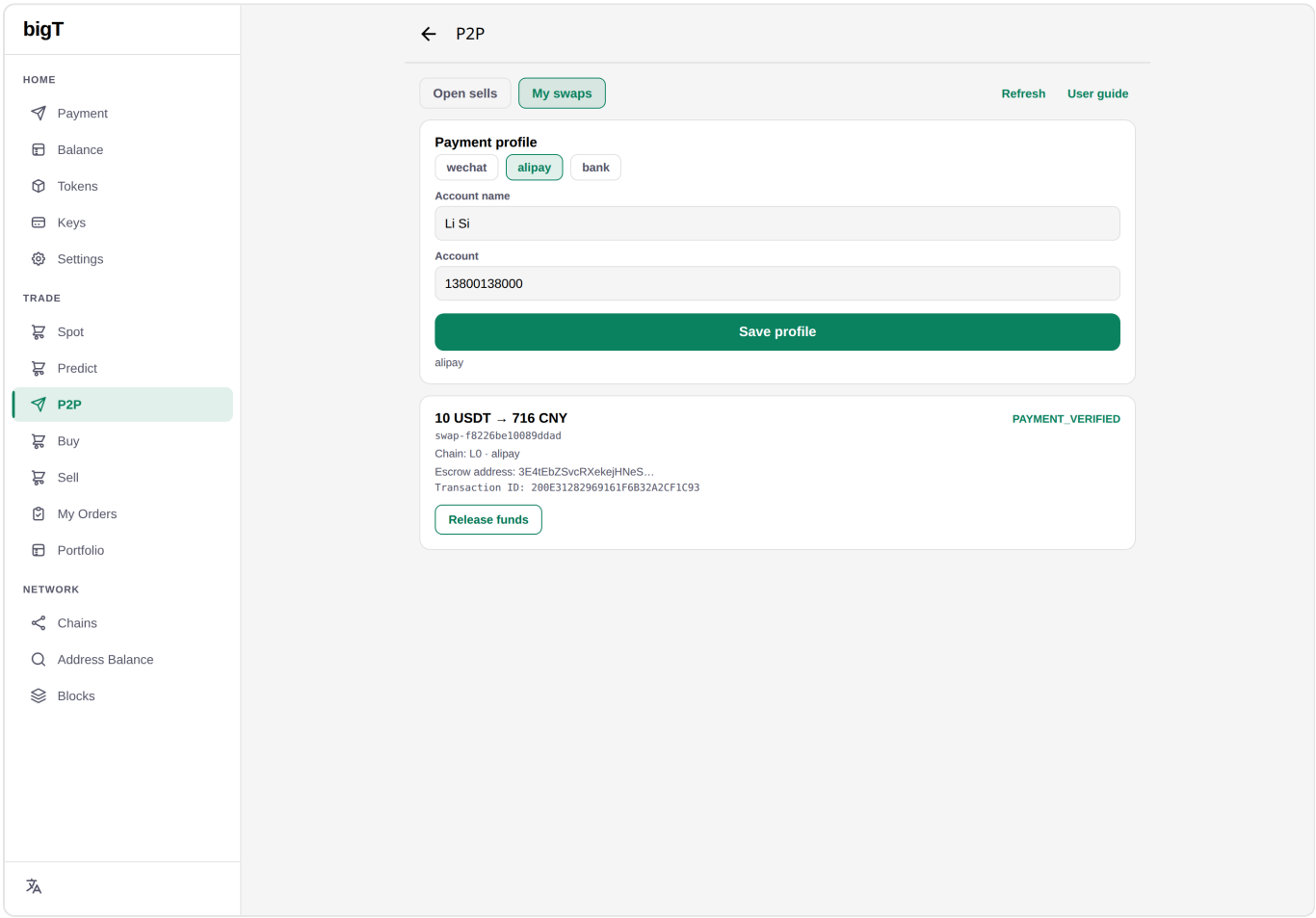
The buyer transfers `716 CNY` in Alipay, pastes the 28-digit trade number (`200E31282969161F6B32A2CF1C93` in this run), and presses **I have paid**.


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HOME
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My swaps
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User guide
Payment profile
wechat
alipay
bank
Account name
Account
Save profile
10 USDT → 716 CNY
PAYMENT_PENDING
swap-f8226be10089ddad
Chain: L0 · alipay
Escrow address: 3E4tEbZSvcRXekejHNeS…
Remark: 91b6eeced27f
Pay the seller
Account name: Li Si
Account: 13800138000
716 CNY
Remark: 91b6eeced27f
Pay by: 10/12/2026, 12:58:45 PM
Transaction ID
200E31282969161F6B32A2CF1C93
I have paid
Dispute

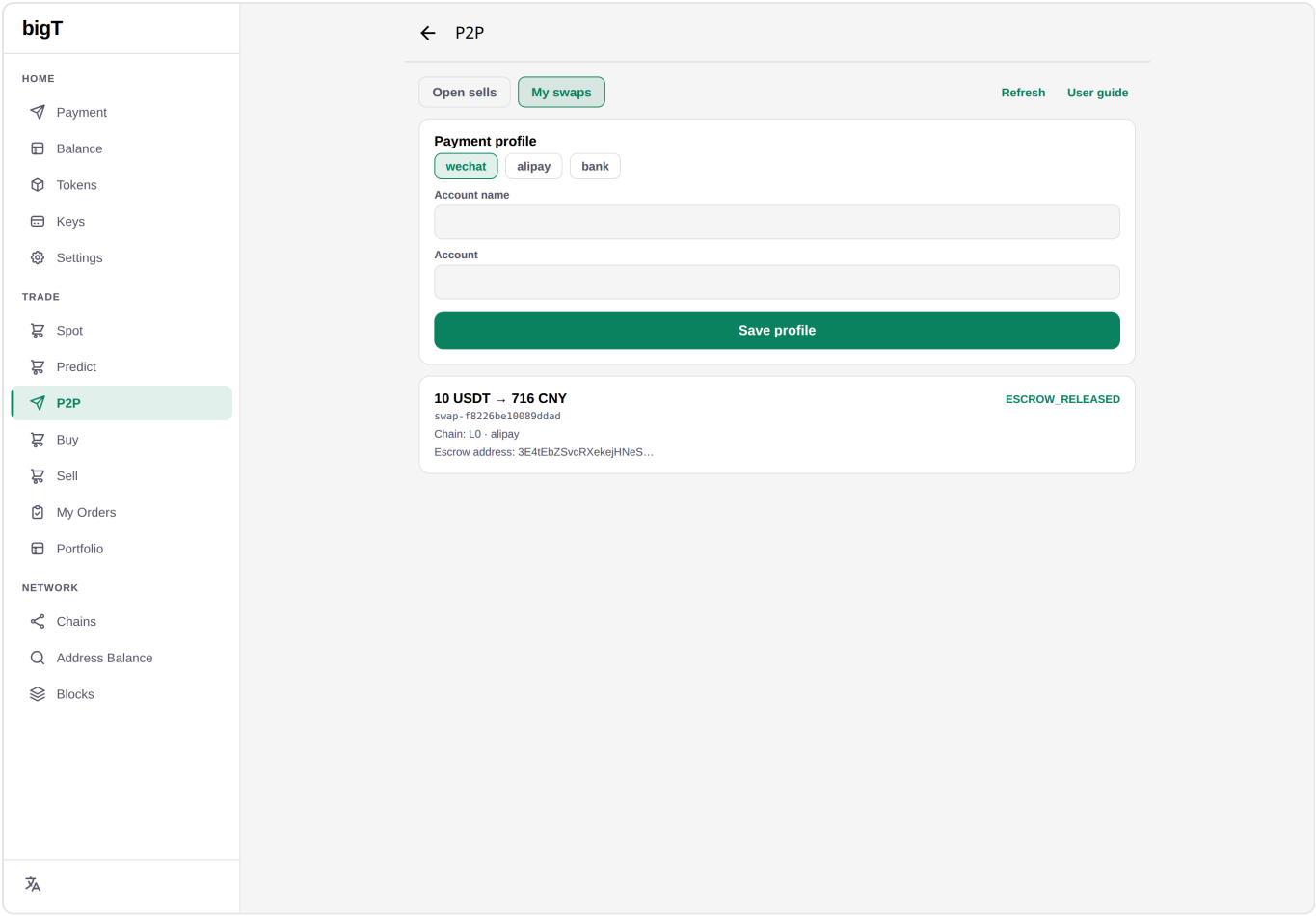

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HOME
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NETWORK
Chains
Address Balance
Blocks
P2P
Open sells
My swaps
Refresh
User guide
Payment profile
wechat
alipay
bank
Account name
Account
Save profile
10 USDT → 716 CNY
PAYMENT_CLAIMED
swap-f8226be10089ddad
Chain: L0 · alipay
Escrow address: 3E4tEbZSvcRXekejHNeS…
Remark: 91b6eeced27f
Transaction ID: 200E31282969161F6B32A2CF1C93
Pay the seller
Account name: Li Si
Account: 13800138000
716 CNY
Remark: 91b6eeced27f
Pay by: 10/12/2026, 12:58:45 PM
Dispute

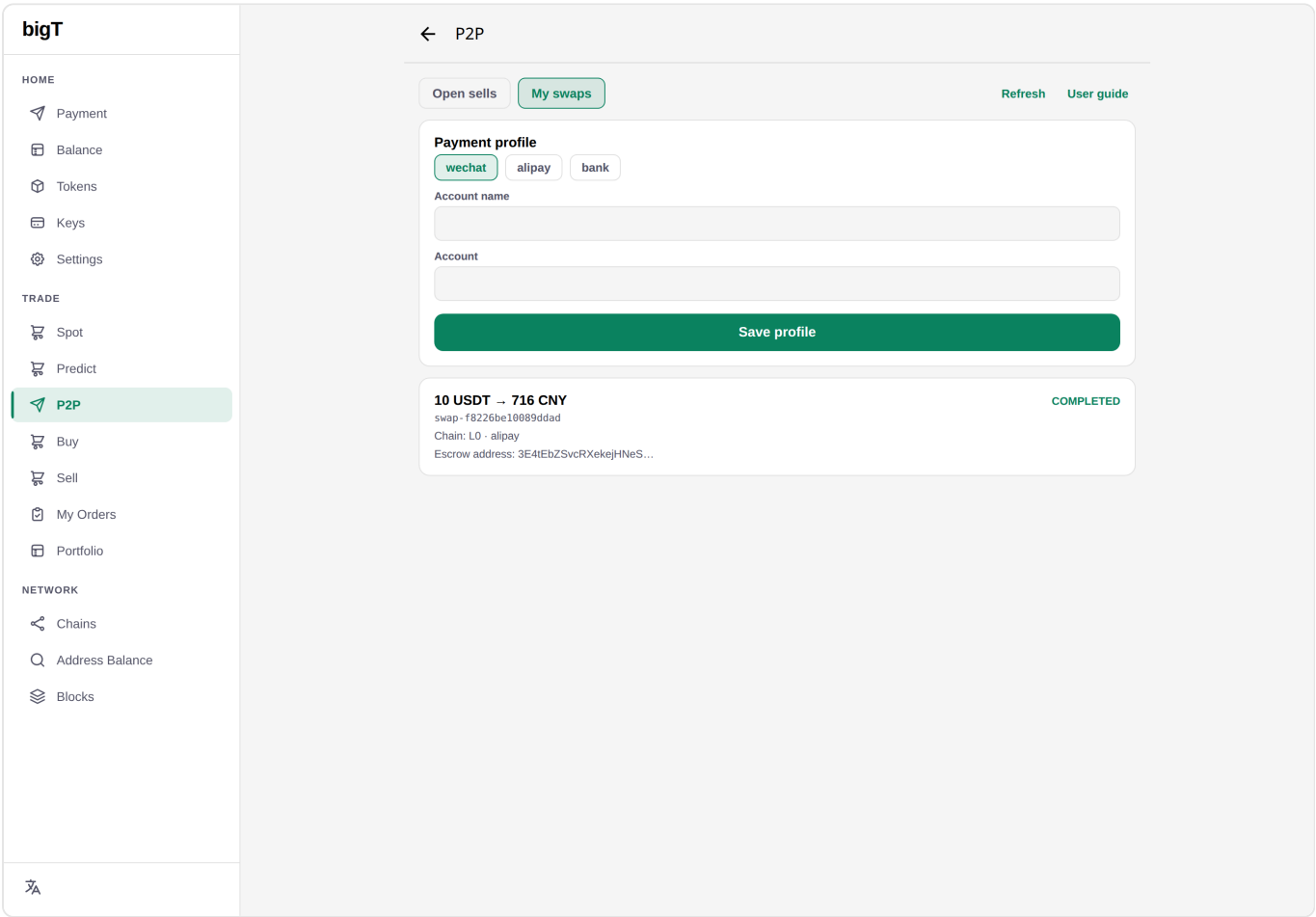
Step 7 — Seller verifies in Alipay, then confirms





Step 8 — Release + complete (engine-signed)

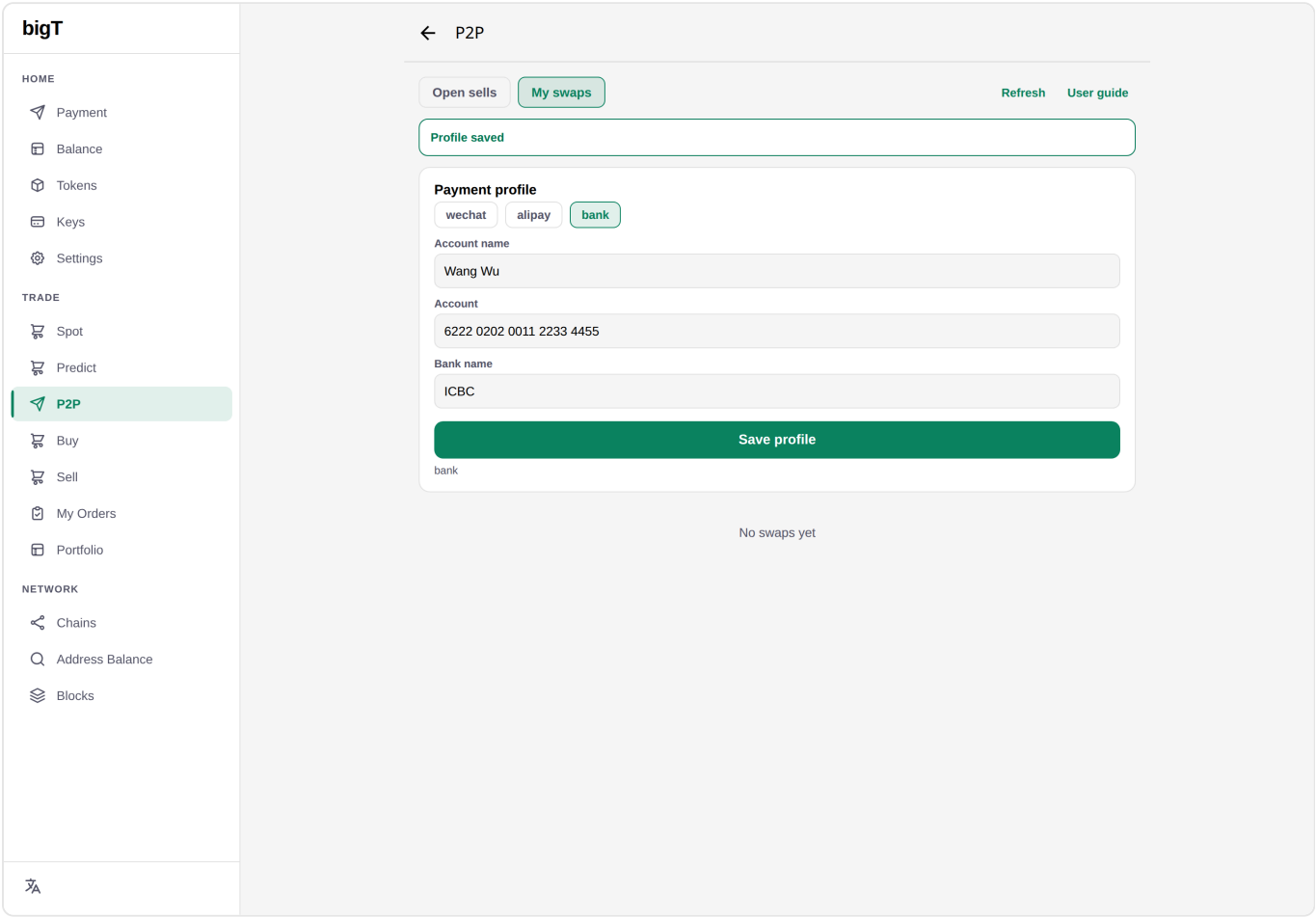




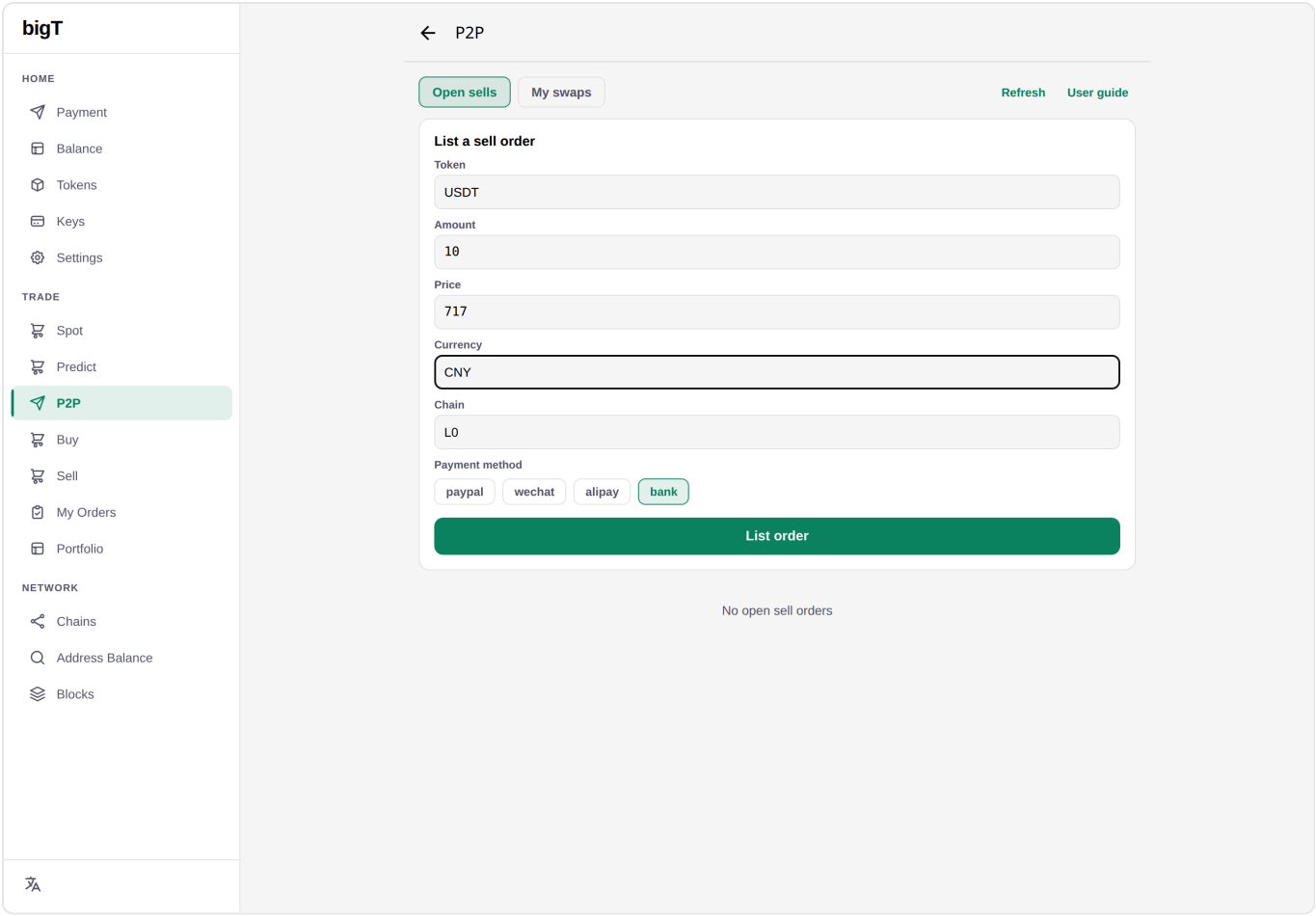
6. Walkthrough — Bank transfer (same shape, bank card)

The bank leg is again the **same state machine** with `wantRail=bank`. The collection profile carries one extra field — the **bank name** — which the instructions panel shows next to the card number. The buyer's 流水号 is the bank's 16-digit transfer reference (starts `3029` in this run).

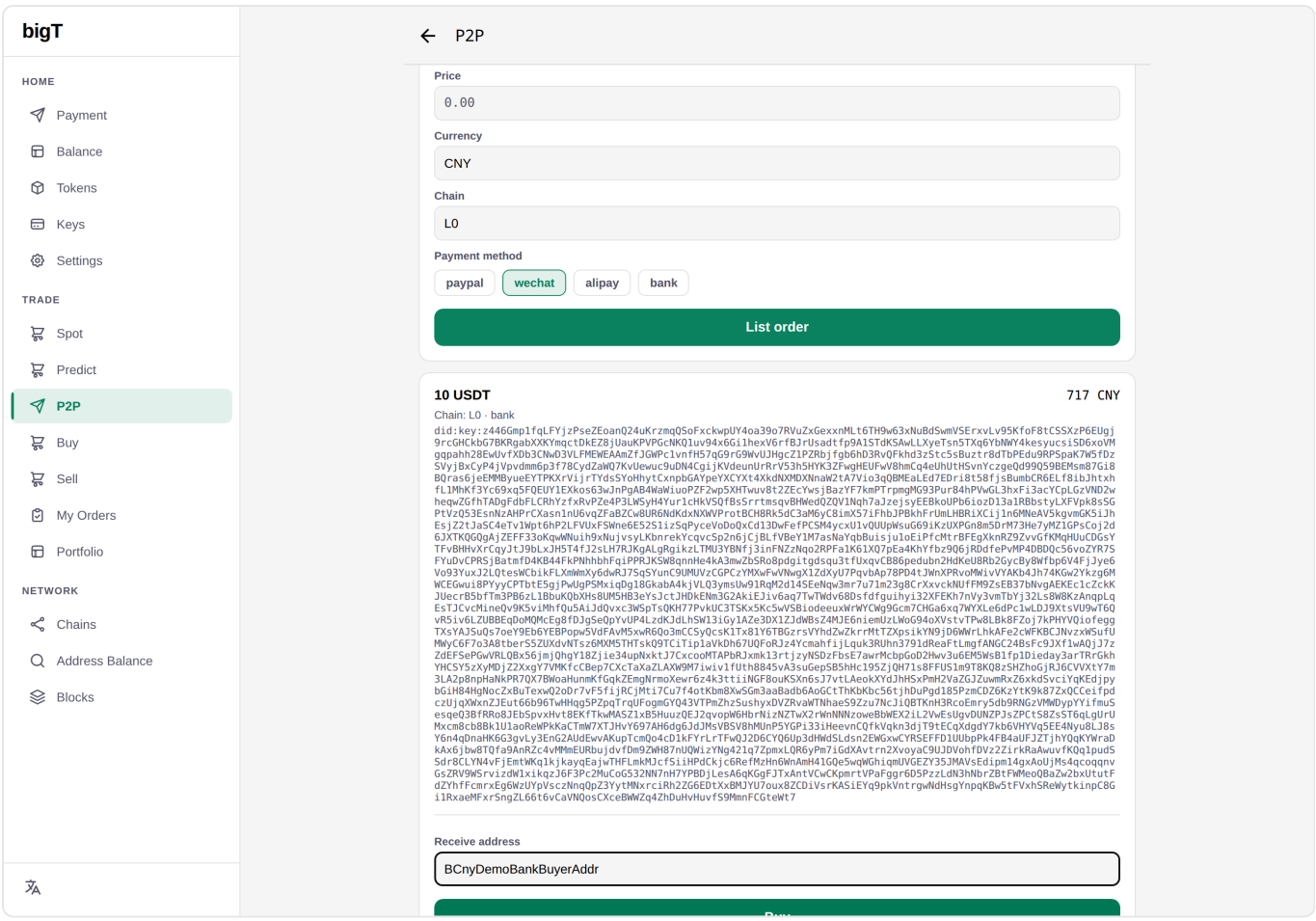
Step 1 — Seller saves the bank profile (name + card + bank)

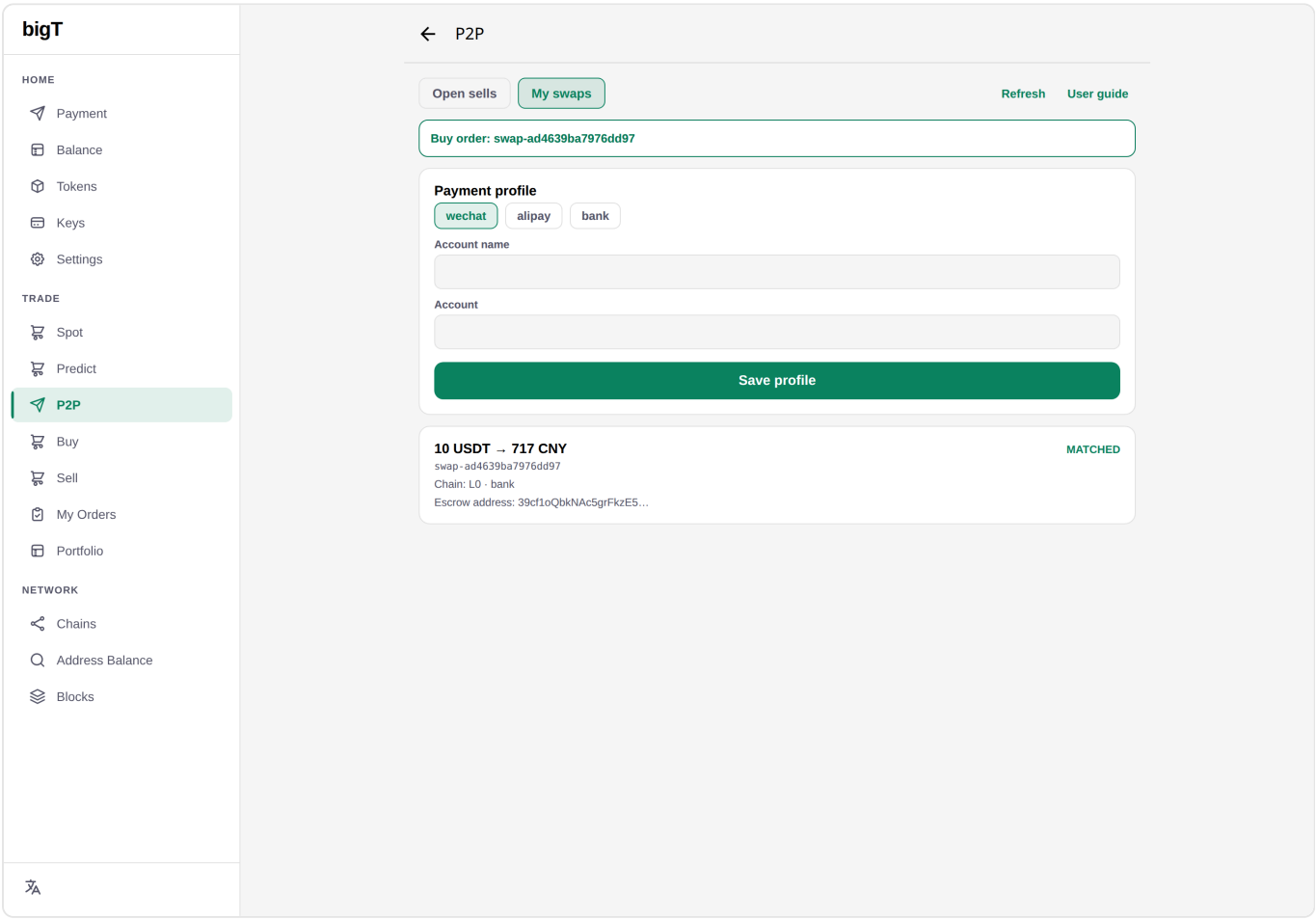


Step 2 — Seller lists 10 USDT for 717 CNY on the bank rail

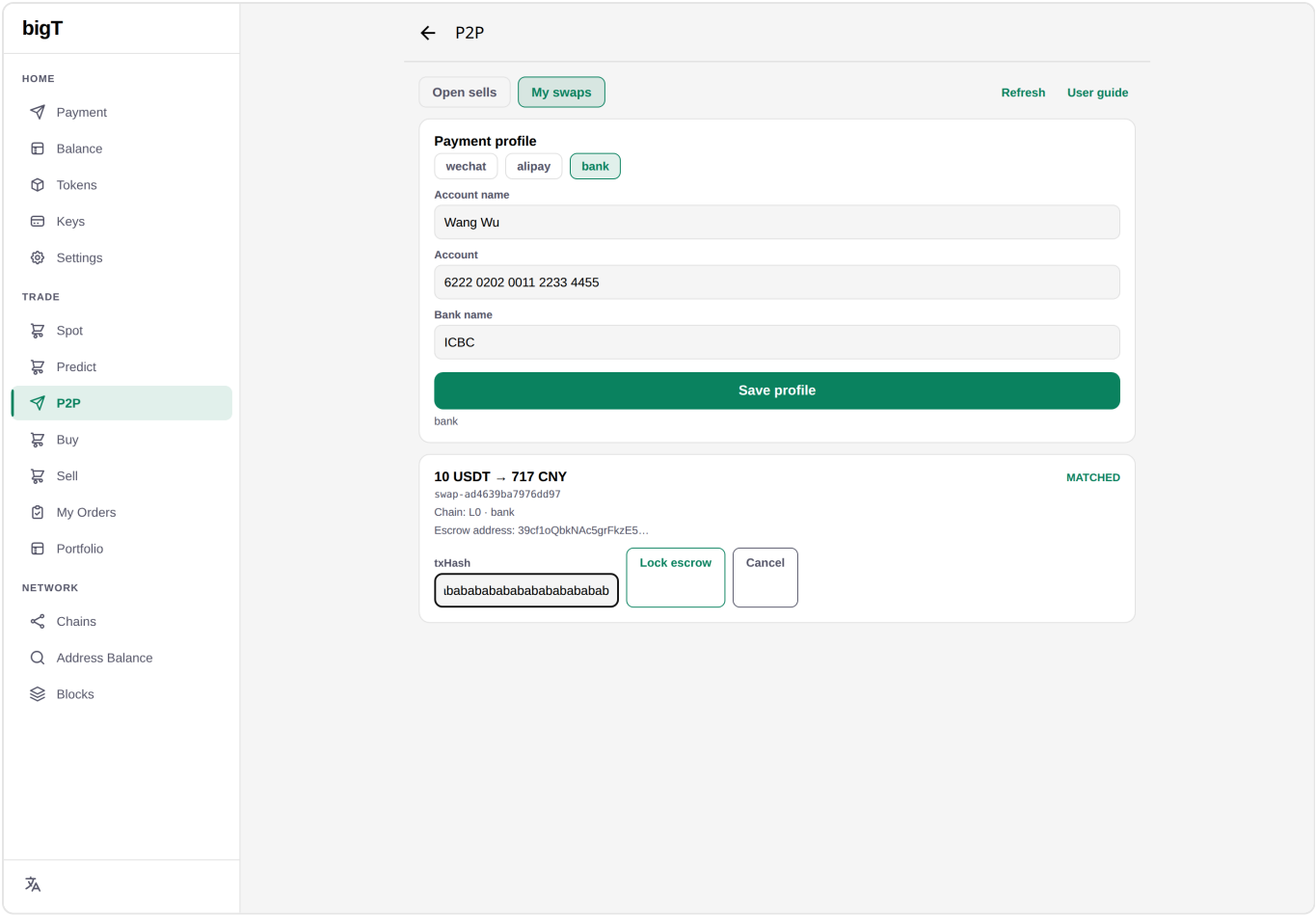


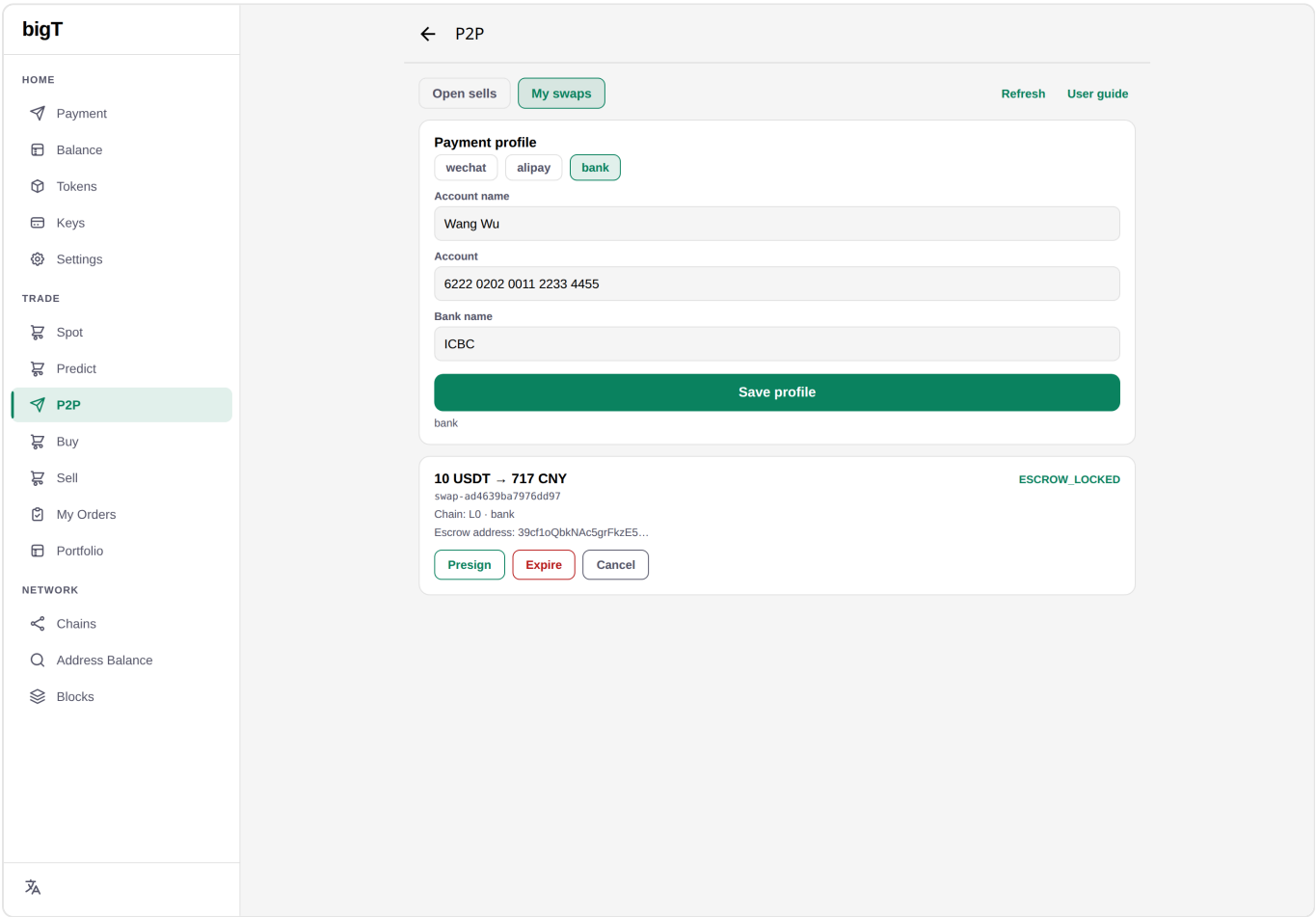
Step 3 — Buyer matches (only a receive address)





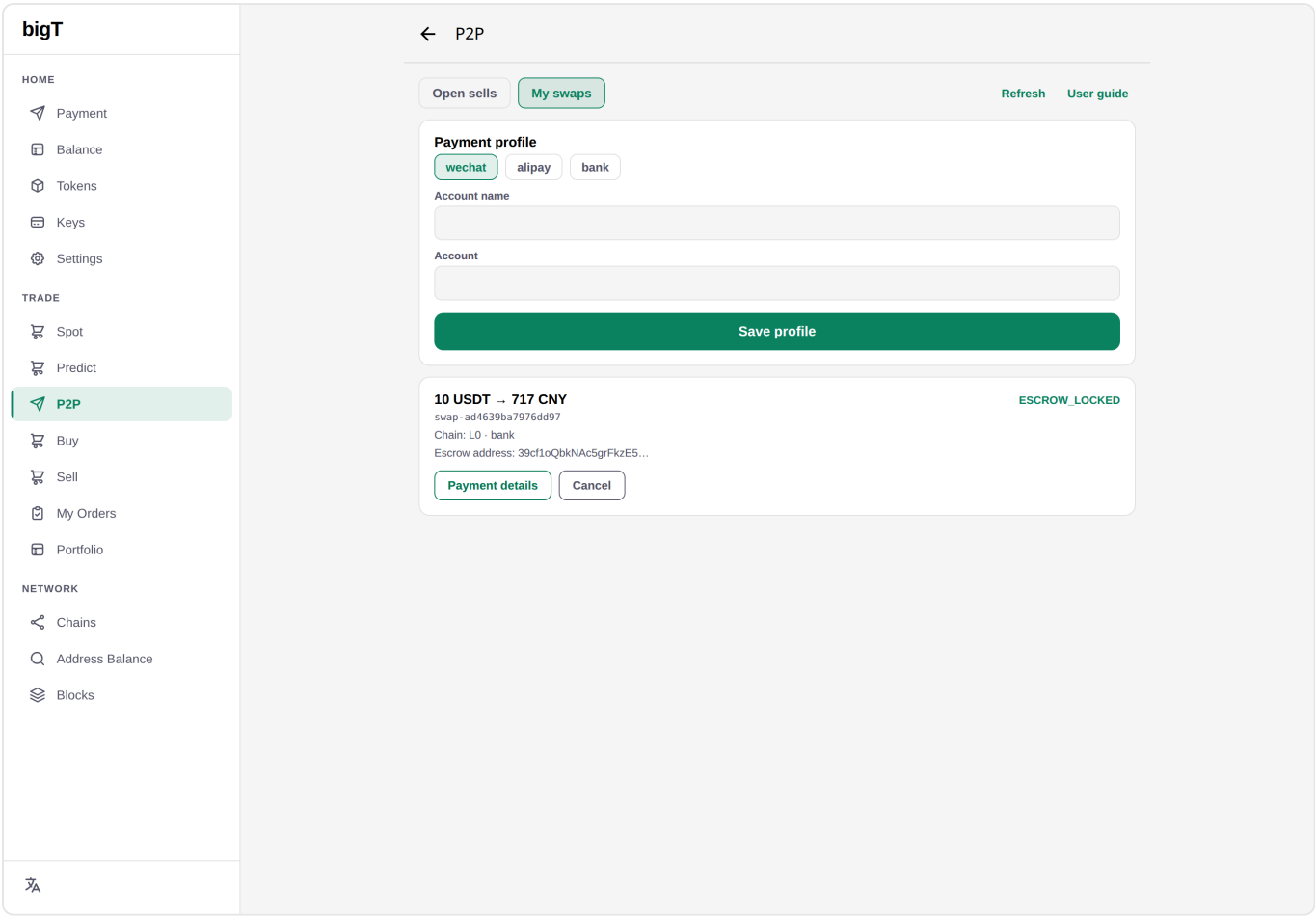
Step 4 — Seller locks the escrow

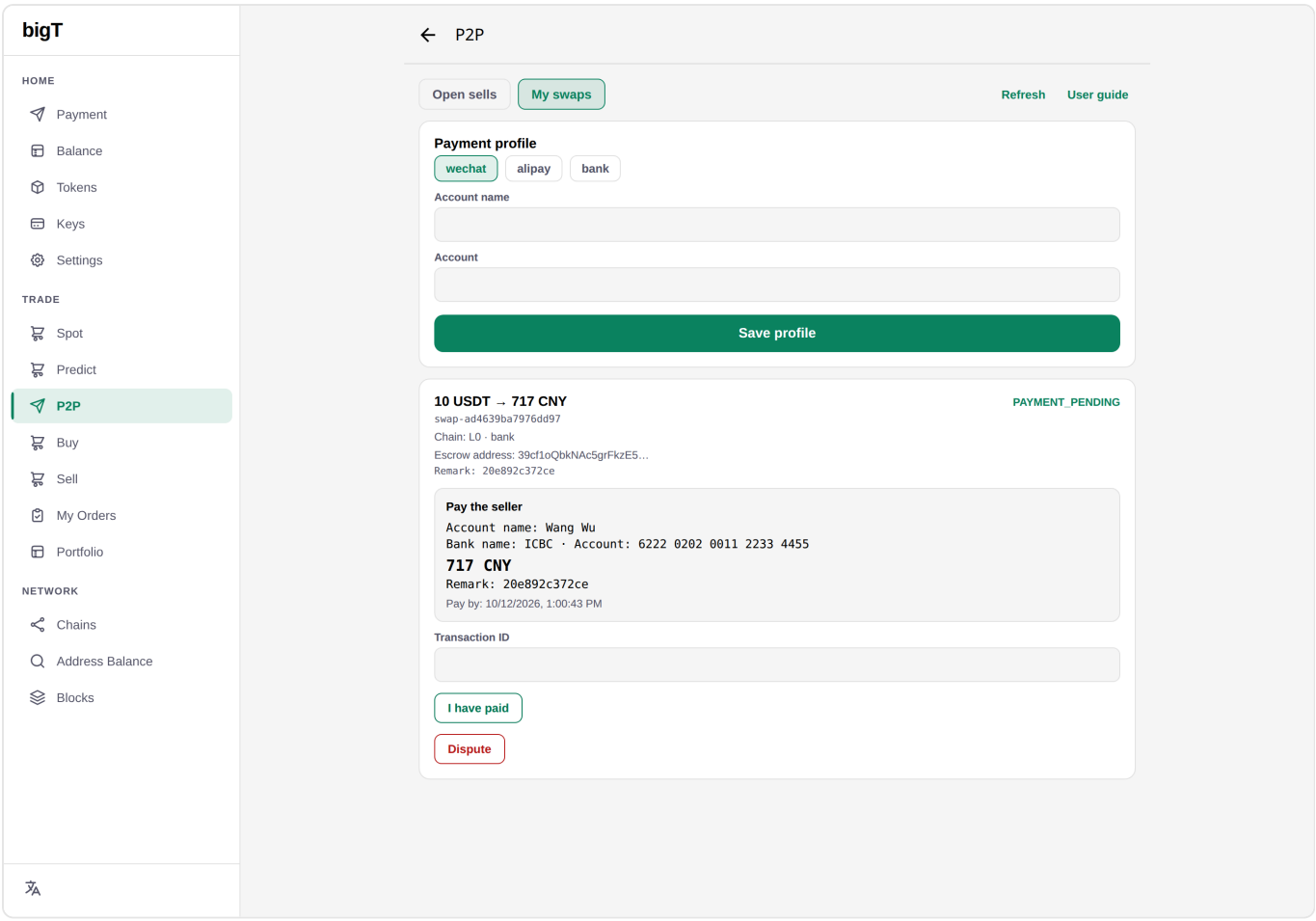




Step 5 — Buyer pulls the bank payment instructions

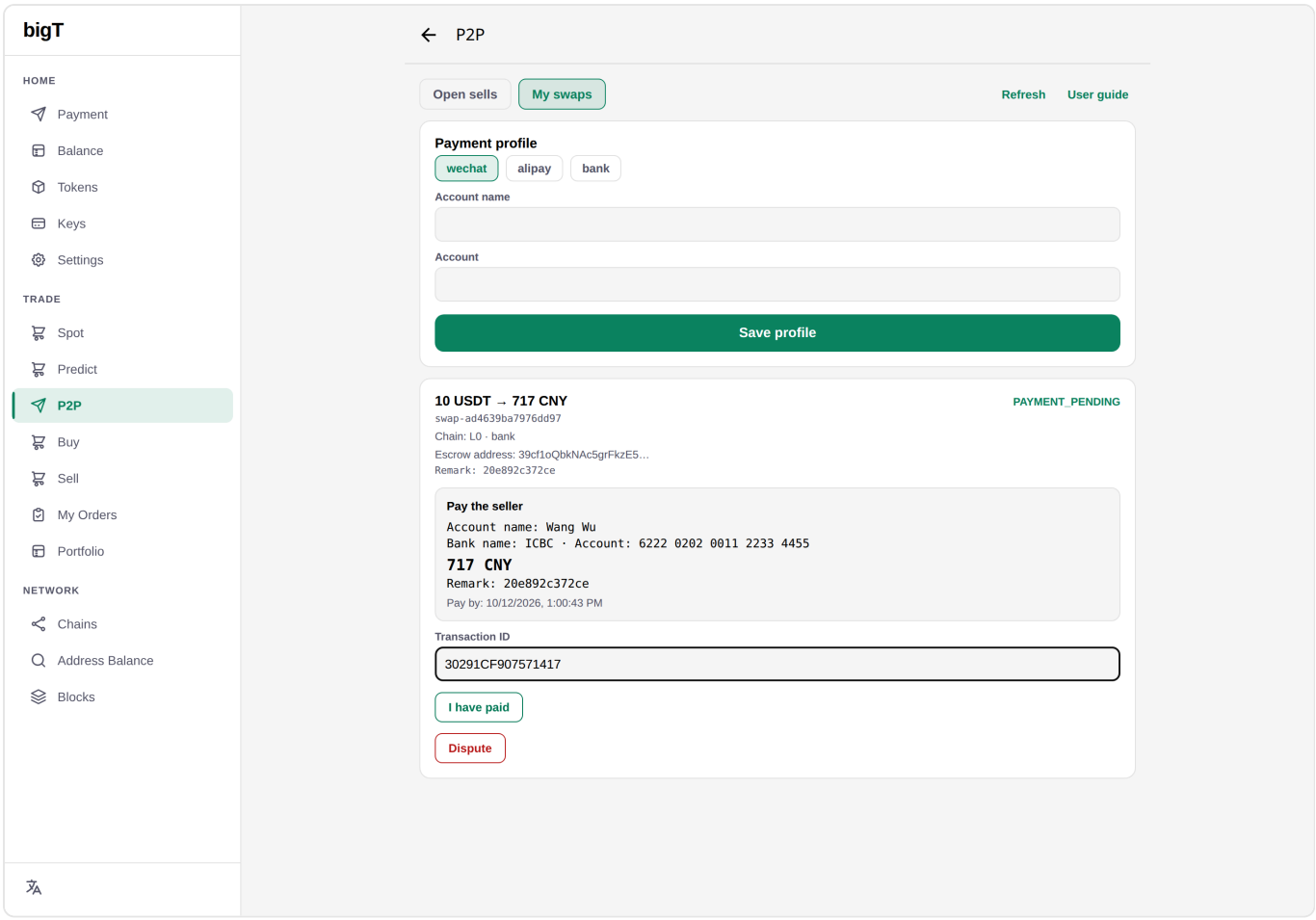

```
{ "method": "bank", "rail": "bank", "accountName": "Wang Wu",
  "bankName": "ICBC", "account": "6222 0202 0011 2233 4455",
  "amount": "717", "currency": "CNY",
  "remark": "20e892c372ce", "payBy": … }
```

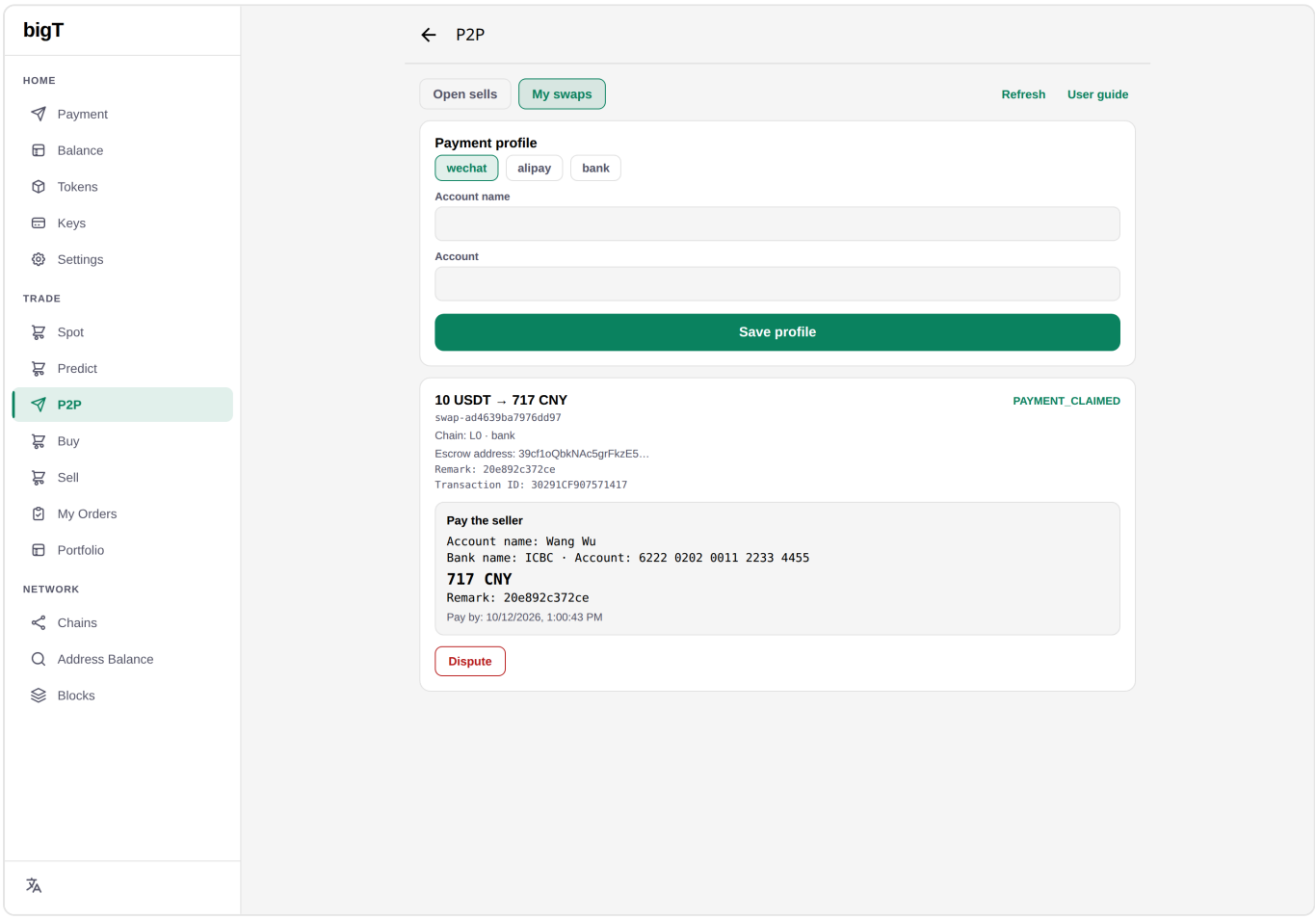





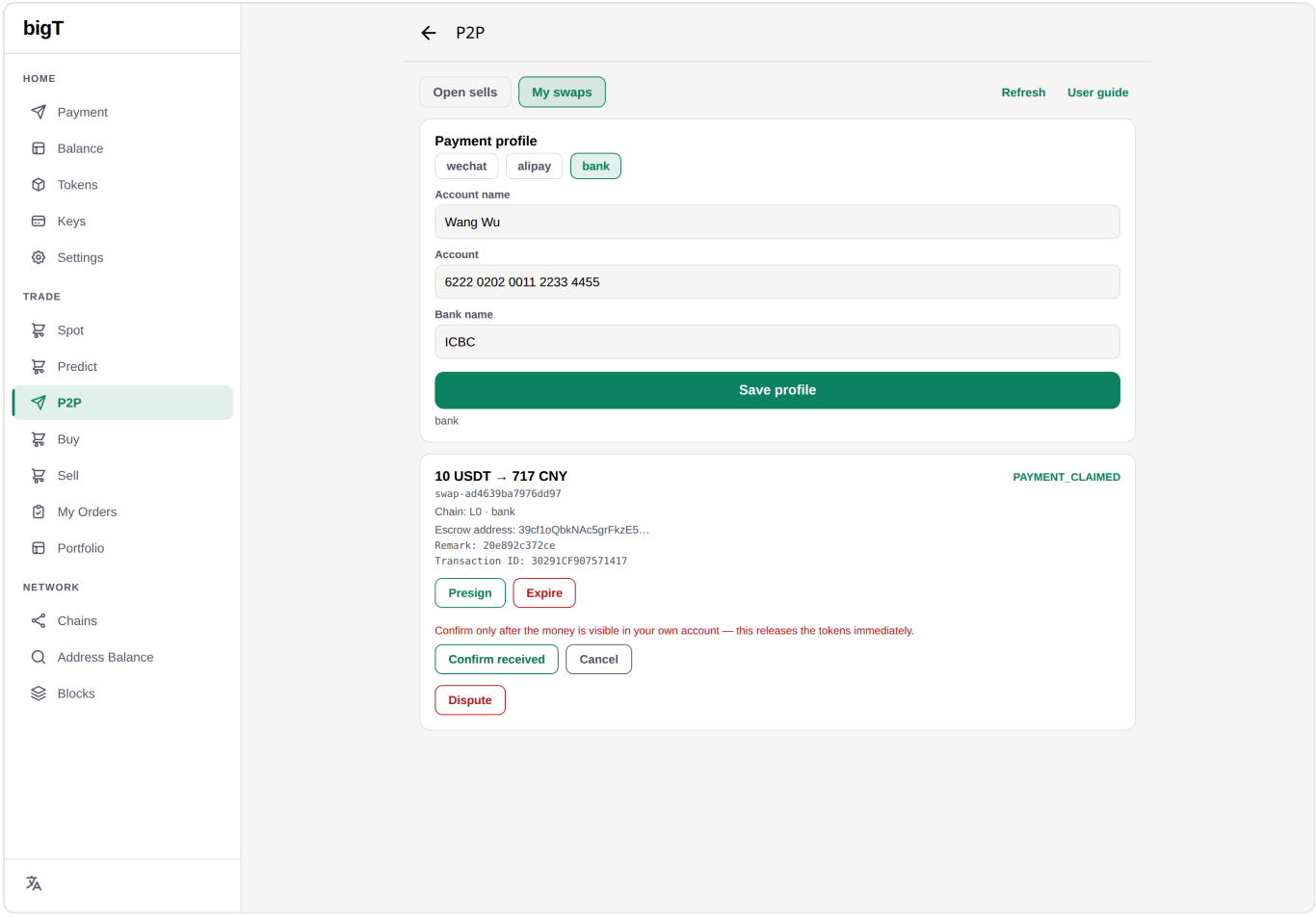
Step 6 — Buyer transfers 717 CNY and claims with the bank reference

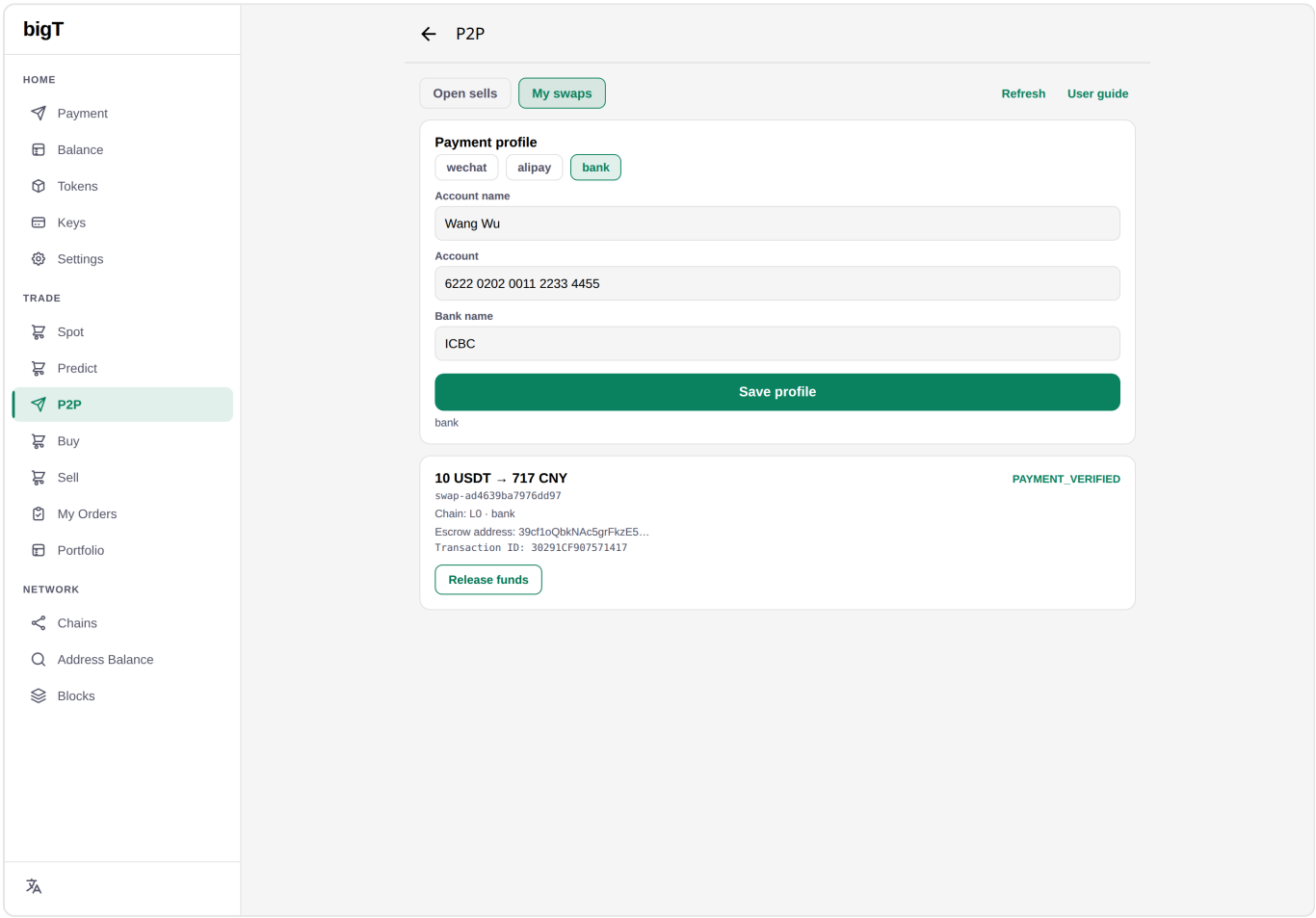
The buyer wires `717 CNY` from their own online banking, pastes the transfer reference (流水号, `30291CF907571417` in this run), and presses **I have paid**.


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Open sells
My swaps
Refresh
User guide
Payment profile
wechat
alipay
bank
Account name
Account
Save profile
10 USDT → 717 CNY
PAYMENT_PENDING
swap-ad4639ba7976dd97
Chain: L0 · bank
Escrow address: 39cf1oQbkNAc5grFkzE5…
Remark: 20e892c372ce
Pay the seller
Account name: Wang Wu
Bank name: ICBC · Account: 6222 0202 0011 2233 4455
717 CNY
Remark: 20e892c372ce
Pay by: 10/12/2026, 1:00:43 PM
Transaction ID
30291CF907571417
I have paid
Dispute

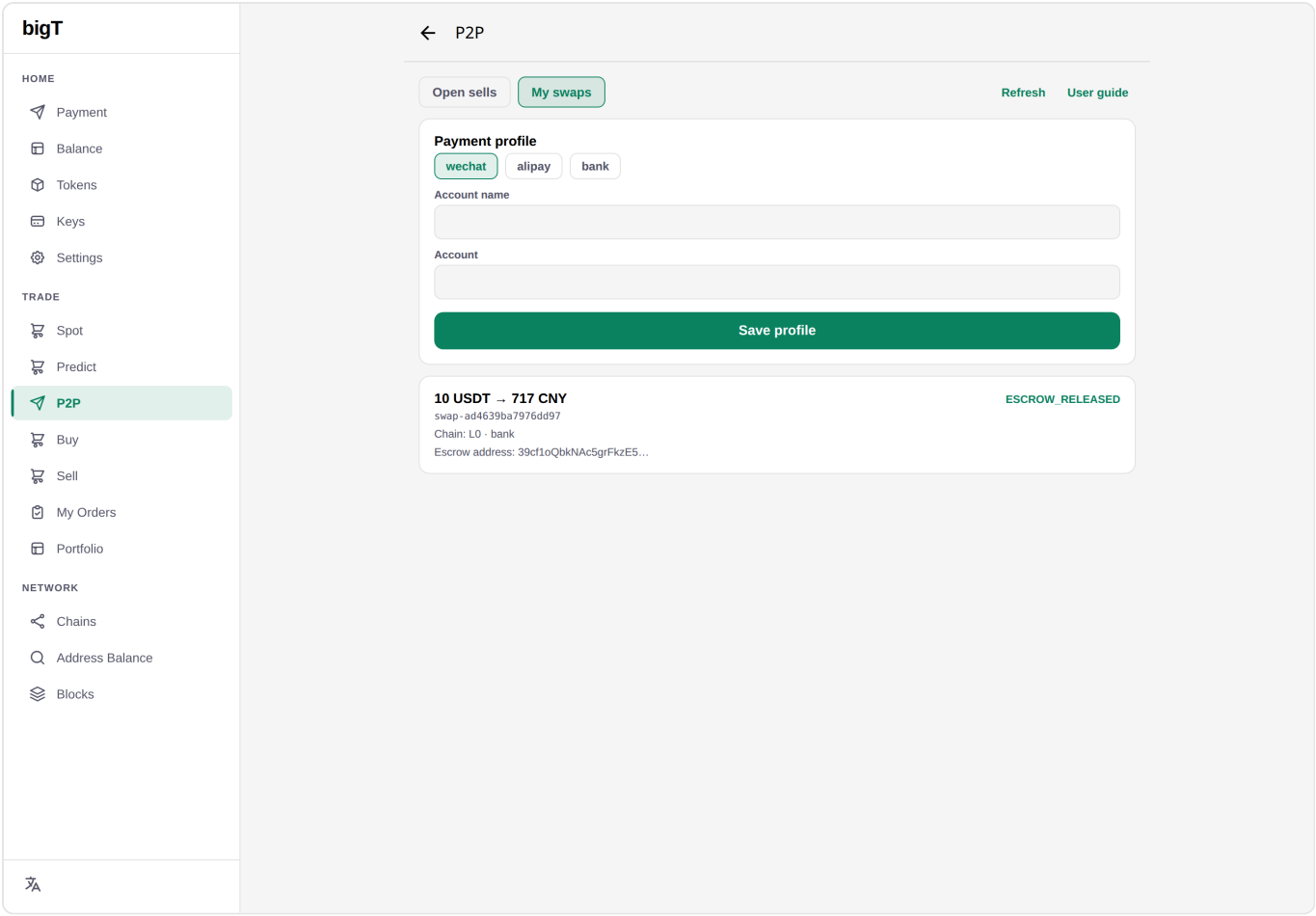

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Blocks
P2P
Open sells
My swaps
Refresh
User guide
Payment profile
wechat
alipay
bank
Account name
Account
Save profile
10 USDT → 717 CNY
PAYMENT_CLAIMED
swap-ad4639ba7976dd97
Chain: L0 · bank
Escrow address: 39cf1oQbkNAc5grFkzE5…
Remark: 20e892c372ce
Transaction ID: 30291CF907571417
Pay the seller
Account name: Wang Wu
Bank name: ICBC · Account: 6222 0202 0011 2233 4455
717 CNY
Remark: 20e892c372ce
Pay by: 10/12/2026, 1:00:43 PM
Dispute

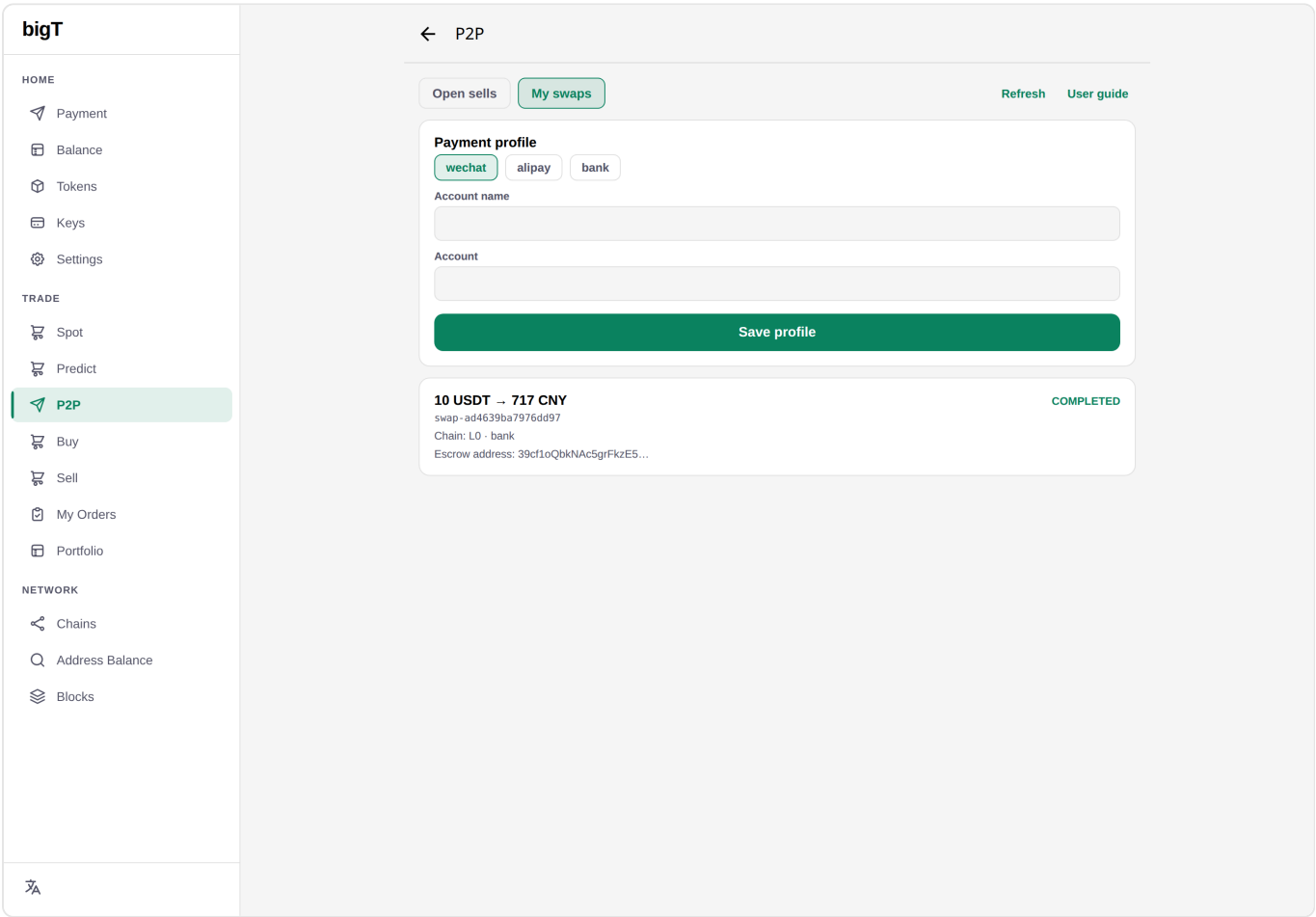
Step 7 — Seller checks their bank statement, then confirms





Step 8 — Release + complete (engine-signed)





7. Timeline of the captured runs

Time	Step
10:40:24	**wechat:** both wallets created
10:40:28	seller payment profile saved (`wxid_pay8888`)
10:40:32	sell order listed (`wantRail=wechat` , 715 CNY)
10:40:35	buyer matched the order (receive address only)
10:41:45	paced 70 s past the rate-limit window
10:41:47	escrow lock recorded (`ESCROW_LOCKED`)
10:41:50	buyer pulled payment instructions (户名 / account / 715 CNY / remark)
10:41:51	buyer claimed payment (流水号 `42034516068EAEC2`)
10:41:54	seller confirmed receipt (`PAYMENT_VERIFIED`) → release → complete
10:42:19	**alipay:** both wallets created
10:42:24	seller payment profile saved (`13800138000`)
10:42:27	sell order listed (`wantRail=alipay` , 716 CNY)
10:43:44	escrow lock recorded (`ESCROW_LOCKED`)
10:43:48	buyer claimed payment (trade no. `200E31282969161F6B32A2CF1C93`)
10:43:52	swap completed (`COMPLETED`)
10:44:17	**bank:** both wallets created
10:44:22	seller payment profile saved (`6222 0202 0011 2233 4455` , ICBC)
10:44:26	sell order listed (`wantRail=bank` , 717 CNY)
10:45:41	escrow lock recorded (`ESCROW_LOCKED`)
10:45:46	buyer claimed payment (bank ref. `30291CF907571417`)
10:45:50	swap completed (`COMPLETED`) — seller already holds the 717 CNY

(The 70 s gap after matching on each rail is deliberate: the engine allows 10 signed requests per DID per 60 s and the seller is at that ceiling after profile + list + match. Full timestamps are in `e2e/demo-output/p2p-cny-capture.json` .)

8. HTTP API reference (CNY rails)

All mutating endpoints take a signed body:

```
{ /* endpoint-specific fields, e.g. action / txId / … */
  "did": "did:key:z446G…",         // ML-DSA-87 did:key — the signer (see table §1)
  "nonce": "b1f0…",                // ≥16 chars, single use
  "timestamp": 1791509100000,      // ms, ±5 min (sign.ts NONCE_MAX_SKEW_MS)
  "signature": "…hex SignatureBundle…" }
```

`signature` = **PQ**: `SHA256(JSON.stringify(payload, sortedKeys))` signed with the did's ML-DSA-87 key, hex-serialized `SignatureBundle` (`sign.ts:39-61` , mirrors the wallet's `p2pIdentity.signP2pPayload`). A classic did instead verifies an Ed25519 signature over the raw bytes, or an ECDSA signature over their sha256 — supported for compatibility, not what new keys use. `payload` is always the body **without** `signature` .

Endpoint	Signer	Required fields	Success
`GET /healthz`	—	—	`{ ok: true }`
`GET /public/orders`	—	—	`{ orders: […] }` (no signature, no PII)
`GET /capabilities`	—	—	`{ rails: […] }` (which rails this engine serves)
`POST /profiles`	seller	`method` , `accountName` , `account` (+ `bankName` , `qr`)	`{ ok, method, updatedAt }`
`POST /profiles/mine`	seller	—	`{ profiles: […] }`
`POST /orders`	seller	`giveToken` , `giveAmount` , `giveChain` , `wantCurrency` , `wantAmount` , `wantRail` , `validUntil`	`201 { orderId, status }`
`POST /orders/:id/match`	buyer	`receiveAddress`	`201 { swapId, status, escrowAddress }`
`POST /swaps/mine`	party	—	`{ swaps: […] }`
`POST /swaps/:id/transitions`	seller (lock/expire/refund) · either (cancel) · **engine** (release/complete)	`action` , + `txHash` for lock/release	`{ swapId, status }`
`POST /swaps/:id/payment-instructions`	buyer	—	`201 { … instructions, remark, payBy }`

`POST /swaps/:id/proof`	buyer	`txId` (1..64 `[A-Za-z0-9._-]`), optional `remark`, `receipt`, `receiptSha256`	`{ swapId, status, txId, receiptSha256 }`	
`POST /swaps/:id/confirm`	seller	—	`{ swapId, status: "PAYMENT_VERIFIED" }`	
`POST /swaps/:id/dispute`	either party	optional `reason` (≤512)	`{ swapId, dispute }`	
`POST /swaps/:id/dispute/resolve`	admin token	`outcome=release\`	refund`	`{ swapId, … }`

Guard rails the engine enforces on every one of them:

- `409` when the status transition is illegal (`state.ts` `VALID_TRANSITIONS`),

`409` on a dedicated action sent to the generic route (`instructions`, `payment_proof`, `payment_confirm` are rejected there),

- `403` unless `did` is the required signer, `400` on bad signature, replayed

nonce, stale timestamp, or `rate limit exceeded` (10/min/DID),

- `422 swap frozen: …` while a dispute is open for the forward actions,
- `400` for rail/action mismatches.

---

9. Review findings (code review of the CNY implementation)

Verified good: the transition table keeps `PAYMENT_CLAIMED` out of the release path; `release` / `complete` are engine-only; the profile is party-scoped; the remark is regenerated per swap and checked on `proof`; the receipt hash is recomputed server-side; unit tests cover the CNY legs (`services/p2p-engine/test/cny.test.ts`).

Open gaps, in the order they matter:

1. **No automatic release after `confirm`.** The engine does not issue `release` / `complete` itself when the seller confirms — an operator/cron must do it with the engine key. Today the wallet UI has no button for it either (by design: the wallet must not hold the engine key).
2. **`payBy` is advisory.** The `SETTLEMENT_CNY_REMARK_TTL` deadline is returned to the buyer but never enforced — a swap can be claimed after it expires, and `SETTLEMENT_CNY_MAX_CENTS` / a confirm timeout are not implemented.
3. **`receiptSha256` hashed the data-URL string, not the image bytes** — fixed: `server.ts` decodes the `…` payload and hashes the

decoded bytes (400 on malformed base64), so the digest is reproducible with `sha256 <file>`; `test/cny.test.ts` asserts it.

4. **`Alert.alert` is a no-op on react-native-web** — every error/success message in the P2P screen (including the payment instructions) was invisible in the web build. Fixed: inline notice banner + inline instructions panel + confirm warning, localized in 12 languages (`p2p.payBy`, `p2p.confirmWarn`).

5. **Rate limit bites scripted flows.** 10 signed calls/DID/minute is tight for the wallet's own read-modify loops (`/swaps/mine` after every transition); the capture script has to pace itself. Worth making `RATE_LIMIT` an env knob if real users ever hit it.

---

10. Regenerating this guide

```
( cd e2e && RAILS=wechat,alipay,bank node capture-p2p-cny.mjs )  # CNY legs (14 shots per rail)
node docs/p2p-demo/scripts/gen-p2p-all-pdf.mjs
# → docs/p2p-demo/assets/p2p-all.pdf
```

Or `./p2pdemo.sh capture walkthrough`. The generator is dependency-free (own markdown → HTML, images embedded as data URIs, `chromium --headless --print-to-pdf`), the same pipeline the dai help guides use.